INCH-POUND

MIL-T-52941B
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SUPERSEDING
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8 May 1985

MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, DIESEL-ENGINE-DRIVEN: PNEUMATIC-TIRED, ROUGH TERRAIN; 4,000 POUND CAPACITY AT 24-INCH LOAD CENTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for diesel-engine-driven, four-wheel drive, rough terrain, pneumatic tired forklift trucks.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

A-A-1108	- Extinguisher, Fire, Vaporizing, Liquid.
GG-P-455	- Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
QQ-P-35	- Passivation Treatments for Corrosion Resisting Steels.
TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Material.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-3859	- Grease Gun, Hand-Operated, Lever, Push and Screw Type.
MIL-P-14105	- Paint Heat Resisting.
MIL-S-40626	- Sign Kit, Vehicle Class.
MIL-S-43926	- Suit, Chemical Protective.
MIL-G-43976	- Glove Set, Chemical Protective.
MIL-F-43987	- Footwear, Cover, Chemical Protective (Overboots).
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty Single Package.
MIL-F-46162	- Fuel, Diesel, Referee.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
MIL-M-51282	- Mask, Chemical - Biological, Field, M17A1.
MIL-H-51291	- Hood, Chemical - Biological Mask: M6A2.
MIL-P-52192	- Primer Coating Epoxy.
MIL-A-52363	- Air Cleaners, Intake: Dry-Type (for Internal-Combustion Engine).
MIL-H-52471	- Hose and Hose Assemblies, Rubber: Hydraulic Pressure-Type, General Specification for.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.
MIL-R-52899	- Ramp Loading Vehicle.
MIL-P-53030	- Primer Coating Epoxy Water Reducible Lead and Chromate Free.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-V-81940	- Valve, Sampling and Bleed, Hydraulic, Type II Systems.

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MIL-T-83133 - Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.
FED-STD-595 - Colors.

MILITARY

MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-162 - Materials Handling Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-248 - Welding and Brazing Procedure and Performance Qualification.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plates Standards.
MIL-STD-1366 - Material Transportation System Dimensional and Weight Constraints, Definition of.
MIL-STD-1410 - Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Material.
MIL-STD-1791 - Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MS24207 - Coupling, Grease Gun; Hydraulic Type Nozzle.
MS51113 - Switch, Vehicular Lights: 24 Volts DC; (Waterproof).
MS51318 - Headlight: Blackout, 24 Volt, Waterproof.
MS51330 - Stop Light - Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS51335 - Pintle Assembly, Towing, 18,000 lbs. Capacity, Manual Release.
MS52125 - Composite Light - Tail, Stop, Turn and Marker.
MS52126 - Composite Light - Front, Turn, Park and Marker.
MS52131 - Connector, Plug, Electrical Intervehicle Power Cable.
MS52149 - Battery, Storage, Lead-Acid, (Low Maintenance).
MS53052 - Bracket Assembly, Liquid Container, Five Gallon.
MS75004 - Adapter, Battery Terminal (Storage Battery).

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MILITARY HANDBOOK

Air Force System Command

Design Handbook DH-11-1	- Air Transportability.
MIL-HDBK-157	- Transportability Criteria.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

TACOM DOCUMENTS AND DRAWINGS

Report No. CR-82-588-003 (Revision 1)	- STE/ICE-R Design Guide for Vehicle Diagnostic Connector Assemblies.
DWG 12258941	- Connector, Receptacle 54 Pin.
DWG 12258955	- Vehicle Test Card.
DWG 12276986	- Kit, M16 Rifle Mounting.

(Application for copies should be addressed to the Department of the Army, U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000

UNITED STATES DEPARTMENT OF AGRICULTURE

Forest Service Standard 5100-1.- Spark Arrestors for Internal Combustion Engines.

(Application for copies should be addressed to Equipment Development Center, 444 East Bonita Avenue, San Dimas, CA 91773.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of issue of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B56.6	Safety Standards for Rough Terrain Fork Lift Trucks.
MH5.1.1M	Requirements for Closed Van Cargo Containers.
MH11.3	Load Handling Symbols for Powered Industrial Trucks.
Z26.1	Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

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(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding and Brazing Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 308 Steel Sheet, Cold Rolled, Long Terne Coated.
D 1171 Rubber Deterioration - Surface Ozone Cracking Outdoors or Chamber.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code Steel, Section 5, Qualification.
D14.3 - Specification for Welding Earthmoving and Construction Equipment, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 550 North LaJeune Road, P.O. Box 351040, Miami, FL 33135.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

Standard 114 - Standard for Break Mandrel Blind Rivets.

(Application for copies should be addressed to the Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

J20 - Coolant System Hoses.
J30 - Fuel and Oil Hoses.
J53 - Minimum Performance Criteria for Emergency Steering of Wheeled Earthmoving Construction Machines.
J98 - Safety for Industrial Wheeled Equipment.
J154 - Operator Enclosures Human Factor Design Considerations.
J164 - Radiator Caps and Filler Necks.
J180 - Electrical Charging Systems for Construction and Industrial Machinery.
J185 - Access Systems for Off-Road Machines.
J198 - Windshield Wiper Systems-Trucks, Buses, and Multipurpose Vehicles.

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J209 - Instrument Face Design and Location for Construction and Industrial Equipment.
J231 - Minimum Performance Criteria for Falling Object Protective Structure (FOPS).
J276 - Steering Frame Lock for Articulated Loaders and Tractors.
J318 - Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers-Trucks, Truck-Tractors, and Trailers.
J350 - Spark Arrester Test Procedure for Medium size Engines.
J386 - Operator Restraint Systems for Off-Road Work Machines.
J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.
J492 - Rivets and Riveting.
J514 - Hydraulic Tube Fittings.
J516 - Hydraulic Hose Fittings.
J517 - Hydraulic Hose.
J518 - Hydraulic Flanged Tube, Pipe, and Hose Connections, 4-Bolt Split Flange Type.
J524 - Seamless Low Carbon Steel Tubing Annealed for Bending and Flaring.
J525 - Welded and Cold Drawn Low Carbon Steel Tubing Annealed for Bending and Flaring.
J534 - Lubrication Fittings.
J536 - Hose Clamps.
J538 - Grounding of Storage Batteries.
J541 - Voltage Drop for Starting Motor Circuits.
J553 - Circuit Breakers.
J585 - Tail Lamps (Rear Position Lamps).
J586 - Stop Lamps for Use on Motor Vehicles Less Than 2032 mm in Overall Width.
J598 - Sealed Lighting Units for Construction, Industrial, and Forest Machinery.
J614 - Engine and Transmission Dipstick Marking.
J645 - Automotive Transmission Terminology.
J678 - Speedometers and Tachometers-Automotive.
J695 - Turning Ability and Off Tracking-Motor Vehicles.
J747 - Hydraulic Control Valve Test Procedure.
J754 - Lubricant Types-Construction and Industrial Machinery.
J833 - USA Human Physical Dimensions.
J847 - Trailer Tow Bar Eye and Pintle Hook/Coupler Performance.
J898 - Control Locations for Off-Road Work Machines.
J899 - Operator's Seat Dimensions for off-Road Self-Propelled Work Machines.
J917 - Marine Push-Pull Control Cables.
J925 - Minimum Service Access Dimensions for Off-Road Machines.
J994 - Alarm-Backup-Electric-Performance, Test, and Application.

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J1024 - Fuel-Fired Heaters-Air Heating-for Construction and Industrial Machinery.
J1040 - Performance Criteria for Rollover Protective Structures (ROPS) for Construction, Earthmoving, Forestry, and Mining Machines.
J1043 - Performance Criteria for FOPS on General Purpose Industrial Machines.
J1117 - Method of Measuring and Reporting the Pressure Differential-Flow Characteristics of the Hydraulic Fluid Power Valve.
J1152 - Braking Performance-Rubber-Tired Construction Machines.
J1163 - Determining Operator Seat Location on Off-Road Work Machines.
J1165 - Reporting Cleanliness Levels of Hydraulic Fluids.
J1176 - External Leakage Classifications for Hydraulic Systems.
J1234 - Specification Definitions-Off-Road Work Machines.
J1308 - Fan Guard for Off-Road Machines.
J1362 - Symbols for Controls, Indicators, and Tell-Tales for Off-Road, and Self-Propelled Work Machines.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

Tire and Rim Association Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Description. The forklift truck (hereinafter called "truck" or "forklift") shall be a four-wheel drive, pneumatic tired, diesel-engine-driven, front loading truck.

3.1.1 Rated load. Rated load (see 6.6.2) for the forklift shall be 4,000 pounds measured with the load center (see 6.6.1) of the weight 24 inches from the forward face of the vertical member of the tines and 24 inches above the load-bearing surface of the forks.

3.1.2 Weight and dimensions.

3.1.2.1 Weight. The weight of the truck shall not exceed 11,000 pounds and shall apply with the forklift fully fueled, serviced, and ready for operation, without operator and load.

3.1.2.2 Overall height. The overall height of the forklift,H1 as defined in SAE J1234, including Rollover Protective Structures (ROPS) and Falling Object Protective Structures (FOPS), shall not exceed 80 inches.

3.1.2.3 Length. The length of the forklift, L1 as defined in SAE J1234, without forks, shall not exceed 166 inches.

3.1.2.4 Width. The width of the forklift, W2 as defined in SAE J1234, including tires, shall not exceed 80 inches.

3.1.2.5 Underclearance. The underclearance of the forklift, including mast, shall not be less than 9 inches and shall apply with the forklift fully fueled and serviced, with rated load in carry position (see 6.6.3). The forklift, with and without load, shall be capable of negotiating a 20-degree ramp with level surfaces at both top and bottom of the ramp. No portion of the forklift, other than the tires, shall come in contact with the ramp when the forklift travels over the ramp in either direction. Hydraulic fittings, hoses, tubing, and mechanical linkages shall not be the lowest portion of the truck or shall be protected by removable structural plates from striking obstacles.

3.1.3 Transportability. The forklift shall be transportable worldwide without damage, by highway, air, marine, and rail transport modes. Transportability criteria are established by MIL-STD-1366 and MIL-HDBK-157 as described herein.

3.1.3.1 Highway transportability. In the forklift's operational configuration, it shall be transportable on a military tractor-trailer combination, without restrictions. The forklift shall be capable of being safely towed behind a M925 truck (5 ton cargo truck NSN: 2320-01-047-8769) with a dead engine on improved roads at speeds of at least 35 mph, for not less than 50 miles, without damage. A towbar shall be furnished and shall be of an A-frame or V-frame design, hinged at two points on the forklift, and with a solid towbar eye in accordance with SAE J847. The towbar shall provide for a turning clearance area in accordance with figure 1. A lockup latch shall be provided to secure the towbar in an upright position when not in use. Two safety chains a minimum length of 1.5 times the towbar length, equipped with safety type grab hooks, shall be furnished; and each chain shall be of an ultimate strength equal to or greater than the weight of the

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forklift. The driveline shall be provided with a means to disengage the wheels from the transmission for towing. Disengagement and reengagement each shall be performed in not more than 15 minutes by the operator using only manual devices and common tools.

3.1.3.2 Air transportability. The forklift shall be transportable in USAF C-130, C-141, and C5A aircraft. Transportability requirements shall be in accordance with MIL-STD-1791, Air Force Design Handbook DH-1-11, and as specified herein. The forklift shall be capable of being driven on and off the aircraft in the operational configuration. The truck shall meet the air transportability requirements with a minimum of 1/2 to a maximum of 3/4 tank of fuel.

3.1.3.3 External helicopter transportability. The forklift, in the operational configuration, shall be capable of being slung (see 3.31) externally under a CH-47 helicopter.

3.1.3.4 Rail transportability. The forklift shall be rail transportable in the continental U.S. (CONUS) and NATO countries without restrictions. The forklift shall have a dimensional profile within the Gabarit International de Chargement (GIC) outline diagram (see figure 2), when loaded on a 50-inch high railcar. The forklift shall be capable of withstanding shock loads resulting from rail impact tests in accordance with 4.6.2.13, without failure, damage, or permanent deformation to any of its components.

3.1.3.5 Marine transportability. The forklift shall be transportable by breakbulk cargo ships, roll-on/roll-off (RORO) ships, C-8 and larger lighter-aboardship (LASH), barge carrying ships (SEABEE), LARC XV and larger amphibious vessels, and Army barges and lighters in accordance with MIL-HDBK-157.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of forklift trucks from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish forklift trucks conforming to this specification.

3.3 Material. Material shall be as specified herein. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the forklifts. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion. Dissimilar metals and methods of protecting them are defined and described in MIL-STD-889.

3.3.1.1 Material deterioration prevention and control. The forklift shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may affect the normal performance of the vehicle or during storage of the vehicle when stored in accordance with proper storage procedures for the type of storage

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involved. Information on the specific material, material finish or treatment for use with components and sub-components shall be made available upon request to the contracting officer or designated representative.

3.3.1.2 Drain holes. Drain holes shall be provided where applicable, to prevent collection or entrapment of water or other unwanted fluids.

3.3.2 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

3.3.3 Tire material. Tires shall not exceed 4 calendar quarters in age from date of cure of the rubber to date of acceptance of the forklift by the Government.

3.3.4 Rubber hose material. Hoses furnished to the Government shall be not more than 12 months old from the date of cure of the rubber to date of acceptance of the equipment by the Government.

3.4 Environmental and use requirements.

3.4.1 Operating conditions. The forklift shall withstand the usage encountered in military operations without damage, permanent deformation, or failure or components. Military operations include loading and unloading the 8 foot wide x 8 foot high family of ASNI/ISO containers at ground level and on chassis, and loading and unloading of flatcars, flatbed trailers, and aircraft with palletized loads of supplies. Operations will be on rough terrain, including unstabilized surfaces such as beaches, snow, mud, and cross country, where high flotation is needed.

3.4.2 Operating temperatures. The engine shall start without preheating within 5 minutes and the forklift shall operate as specified herein within 15 minutes after engine start in any ambient temperature in the range of -25 °F to +120 °F.

3.4.3 Storage temperature. The forklift shall not be damaged by storage under the conditions indicated in table I.

TABLE I. Storage.

Air Temperature	Time Period
+160 °F	1 hour daily.
+155 °F	4 hour daily.
-25 °F	6 hour daily.

3.4.4 Rain. The forklift shall start within 5 minutes and operate as specified herein when tested as specified in 4.6.2.28:

a. Without malfunction of any electrical component, light, gauge, or instrument.
b. Without leakage of water into the air intake filter or canister.

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c. Without leakage of water into the torque converter oil chamber, engine crankcase sump, transmission, hydraulic system, brake fluid system, transfer case, instruments, gauges, or fuel tank.
d. Without leakage into the cab, when furnished.
e. Without leakage into the toolbox.

3.5 Safety. The forklift shall meet the safety requirements specified in SAE J98 and ANSI B56.6. All rotating and reciprocating parts, parts energized electrically, and parts subject to high temperatures shall be guarded when such parts are exposed to contact by operator and maintenance personnel performing daily operating and maintenance functions. Protective devices shall not impair the operating functions. All heat exchangers shall be protected from damage by location or by a grilled guard. Steps and platforms for entering and servicing the truck shall have antiskid surfaces. For forklifts with hinged-frame (articulated) steering, a warning notice shall be stenciled near the hinge on both sides of the forklift, stating "WARNING, DO NOT STAND HERE", or words to that effect (see 3.32). For forklifts with Ackermann type steering, a warning notice shall be stenciled behind the rear wheel on both sides of the forklift, stating "WARNING, DO NOT STAND HERE", or words to that effect (see 3.32). Nonfunctional sharp edges, projecting points, and excessive length of fasteners shall be avoided. Oil level indicators for engine, transmission, and other components shall be within easy and safe access of the operator or maintenance personnel.

3.5.1 Rollover protective structures (ROPS) and falling object protective structures (FOPS). The forklift shall be equipped with a complete ROPS and FOPS structure. The ROPS conforming to SAE J1040, and FOPS conforming to SAE J1043, or SAE J231. Installation and removal of the ROPS and FOPS shall be accomplished without cutting or welding. Permanent or removable lifting eyes/devices shall be provided to facilitate removal of the ROPS and FOPS. The ROPS and FOPS shall be free from sharp edges, particularly at any point where a sling may come in contact during lifting (see 3.31). Quick disconnects for electrical lines shall be provided to facilitate removal of ROPS and FOPS and cab when specified (see 6.2).

3.5.2 Seatbelt. An adjustable, weather resistant seatbelt conforming to SAE J386 shall be provided with a corrosion resistant, metal-to-metal, quick release buckle. The seatbelt shall be furnished equipped with sleeves and retractors or seat mounted.

3.5.3 Backup alarm. A backup alarm conforming to SAE J994, type C or D shall be furnished. The backup alarm signal shall be audible above, and distinguishable from, the surrounding vehicle noise level. The backup alarm shall be automatically disengaged in the blackout mode.

3.5.4 Fire extinguisher. A Coast Guard approved bracket shall be mounted inside the operator's cab or in a suitable location. The bracket shall secure a 5 pound Halon 1211 fire extinguisher conforming to A-A-1108, type B. When specified (see 6.2), the fire extinguisher shall be provided.

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3.5.5 Mirrors. Mirrors shall be provided within the outline of the forklift to provide for operator viewing of the area immediately behind the forklift without the operator looking back. One mirror shall be provided on each side of the forklift.

3.6 Human factors engineering. The characteristics of the forklift, with ROPS and FOPS installed, shall provide for operation and maintenance by personnel ranging from the 5th percentile female clothed through the 95th percentile male arctic clothed, in accordance with SAE J833, SAE J154, SAE J925, and MIL-STD-1472 as specified herein. The operator shall be able to see the tip of at least one fork at any lift height without leaving his seat when there is no load on the forks. With forks in the carry position, the cross members of the mast structure shall not interfere with the operator's line-of-sight between horizontal and 10 degrees below horizontal using the range of eye heights above the operator's seat as specified in SAE J833. As a minimum each front wheel and the left rear wheel of the truck shall be directly visible to the operator while the operator remains in the seat in position to control the truck's drive controls.

3.6.1 Noise limits. The noise produced by the forklift shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474 paragraphs 5.2, 5.3, and 5.4, when tested in accordance with 4.6.2.27. The provision of MIL-STD-1474, paragraphs 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, paragraph 5.1.1.2 procedures have been pursed and documented to the satisfaction of the procuring activity not later than one month prior to first article test and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, paragraph 4.3, except the sign will state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET." The sign shall be readable at the distance indicated on the sign.

3.7 Maintainability. Provisions shall be made for adjustment, servicing, or replacement of parts and components. Items requiring periodic or preventative maintenance such as filters, belt tensioning devices, drain plugs, etc., shall be accessible without requiring removal of other components. When access openings are used on the exterior of the forklift, the edge of the opening shall be smooth and shall be provided with a removable or hinged cover whenever it is required. Dimensions of hand or arm access opening shall be in accordance with SAE J925 for arctic clothing. When tested as specified in 4.6.2.36, the forklift shall have a maintenance ratio (MR) of not more than 0.15. The MR is defined as the ratio of total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components are excluded. Scheduled maintenance shall be required not more often than every 50 hours, except for before and after operation checks. Each operation in the following maintenance operations list shall be accomplished by one person (except where otherwise specified) in not more than the time specified, using common tools:

a. Remove, replace, and adjust all engine-driven belts - 3/4 hour.
b. Remove and replace alternator - 1/2 hour.
c. Remove and replace all hydraulic-system filters, screens, and strainer - 1 hour.

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d. Remove and replace engine coolant-system hoses - 1 hour.
e. Drain engine lubricating oil, remove, and replace oil filter elements and refill crankcase - 1 hour.
f. Remove and replace fuel filter elements - 1/2 hour.
g. Disconnect battery cables, remove and replace batteries, reconnect battery cables - 1/2 hour.
h. Drain torque converter oil and transmission oil, remove and replace all transmission filter elements and strainers, and refill converter and transmission - 1 hour.
i. Remove and replace starter - 1 hour.
j. Bleed and adjust brakes and refill master cylinder (if applicable) - 2 persons - 1 hour.
k. Remove and replace floodlight and taillight bulbs (not including blackout lights) - 1/4 hour each bulb.
l. Remove, disassemble, service, and install air cleaner - 1/2 hour.
m. Lubricate all lubrication fittings with 2 strokes per fitting using the gun and extension specified in 3.29.1 - 1 hour.

3.8 Reliability. The specified mean-time-between-hardware-failures (MTBHF) shall be 75 hours when tested in accordance with 4.6.2.35. The specified mean-time-between-unscheduled-maintenance-action (MTBUMA) shall be 15 hours when tested in accordance with 4.6.2.35.

3.9 Performance. The forklift shall conform to the requirements specified herein under all conditions specified without malfunction, damage, or permanent deformation. There shall be no evidence of accelerated wear or failures when operated under all conditions specified herein.

3.9.1 Stability.

3.9.1.1 Static. The forklift shall conform to ANSI B56.6 stability requirements.

3.9.1.2 Dynamic. The forklift shall negotiate a 30 percent slope with and without rated load in the centered carry position (see 6.6.3) in full circle operation at maximum steer angles in both directions without any wheel leaving the ground. The forklift while descending a 30 percent grade in the forward direction with rated load in the carry position at not less than 2 mph, shall stop by applying the service brake without the rear wheels leaving the ground.

3.9.2 Vehicle floatation. The forklift shall have a flotation index (FI) of not more than 25, with and without rated load on the forks in the carry position (see 6.6.3). The FI shall be calculated using the following formula:

$$FI = \text{Antilog} \left[\frac{0.211 - 0.35\,C + 0.0071\,P - 0.0752(P_F - P_R)}{P} - \frac{-0.0402(AF - AR) + 0.0346\,K}{A} \right]$$

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Where: P_F = Tire pressure, front (psi).
P_R = Tire pressure, rear (psi).
P = Tire pressure, average = ½ (P_F + P_R).
A_F = Axle load, front (lbs).
A_R = Axle load, rear (lbs).
A = Gross Vehicle weight = A_F + A_R.
K = 53.75 + 0.000139A (for nonradial tires).
K = 55.77 + 0.000032A (for radial tires).

$$C = \log \left[\frac{A}{0.607P + 1.35 \left[\frac{117R}{5W + D} \right] - 4.93} \right]$$

R = Tire ply rating.
W = Tire width (in - as cited in tire size).
D = Rim diameter (in - as cited in tire size).

The FI shall be determined with the forklift fully fueled and serviced with lubricants. For the purpose of calculating both loaded and unloaded FI, the tire inflation pressures may be reduced by not more than 30 percent of the recommended pressure published in the TRA Yearbook or tire manufacturer's, published literature for the forklift's maximum tire load at a 5 mph rating.

3.9.3 Longitudinal gradeability. The forklift shall ascend a 45 percent grade in forward gear range at a speed of not less than 2 mph with rated load.

3.9.4 Travel speed. The forklift shall be capable of controlled travel speeds of not less than 20 mph forward and not less than 10 mph in reverse on a level surface with and without rated load.

3.9.5 Fording. The forklift shall operate in water to a depth of 20 inches. Forklift components shall be protected from damage, malfunction, and corrosion due to immersion followed by drying for those components below the fording level, and due to splash and spray for those components above the fording level. No operating controls nor the operator's seat cushion shall be lower than the fording level. When tested as specified in 4.6.2.31, for a minimum of 2.5 hours the truck shall operate during and after fording operation without malfunction or damage from corrosion.

3.9.6 Load transfer. When tested as specified in test method No. 31 in a simulated container/trailer loading operation, the forklift shall transfer rated loads at a rate of not less than 1 cycle per 1.2 hours.

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3.9.7 Lifting speed. Speed of lift with rated capacity load shall be not less than 60 feet per minute over the entire distance from ground level to maximum fork height.

3.9.8 Lowering speed. Speed of lowering of unloaded forks shall be not less than 40 feet per minute over the entire distance from maximum fork height to ground level at engine idle. Speed of lowering forks with rated load shall be not more than 80 feet per minute over the entire distance from maximum fork height to ground level at engine idle. The hydraulic control system shall be damped, metered, or easily manually regulated to reduce shock and prevent the truck from overturning when rated load is lowered at maximum speed and then stopped quickly.

3.9.9 Curb clearance circle. The curb clearance circle when measured in accordance with the Field Test Procedure in SAE J695 shall not be greater than 28 feet when carrying rated load in the carry position for both the forward and reverse directions.

3.9.10 Upright tilt. When unloaded, truck mast shall have 10 degrees minimum forward tilt and 20 degrees minimum rearward tilt. Positive means shall be furnished to prevent cavitation of the tilt cylinders at a minimum of 120 °F oil temperature when tilting rated load to full forward tilt at engine low idle speed. Maximum time to tilt from full rear tilt to full forward tilt at maximum fork height shall be 30 seconds at low idle engine speed. The truck shall tilt rated load at maximum fork height from full forward tilt to full rear tilt in no more than 10 seconds at governed engine speed.

3.9.11 Collapsed mast height. The collapsed mast height of an unloaded truck measured from the ground to the top of the uprights in a vertical position shall not exceed 78 inches.

3.9.12 Maximum fork height. The fork height of a truck bearing rated capacity load in the horizontal position and the uprights fully extended, measured from the ground to the load carrying surface of the forks at the rated load center, shall be not less than 120 inches.

3.9.13 Free lift height. Height of forks without increase in the specified collapsed mast height shall be not less than 48 inches with rated load.

3.9.14 Low engagement height. The lowest engagement height of the forks shall be not less than 4 inches below ground level when measured at the load carrying surfaces of the forks on an unloaded truck with uprights in a vertical position.

3.9.15 Drift. The lift assembly shall be capable of holding the rated load at maximum height for not less than 2 minutes with not more than 1 inch of vertical drift and not more than 0.5 degree of rotational drift from the vertical. Specified drift rates shall be measured when hydraulic oil temperature is not less than 120 °F.

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3.9.16 Oscillation. Hydraulically powered oscillation of the forks shall be furnished. The forklift shall be capable of oscillating rated load at each maximum side shifted position. The oscillation of the rated load from horizontal shall be not less than 10 degrees clockwise when the load is side shifted to extreme left and not less than 10 degrees counterclockwise when load is side shifted to extreme right (20 degrees total oscillation). Time to oscillate the total 20 degrees shall be not more than 7 seconds at low engine idle and not more than 3 seconds at high engine idle for both clockwise and counterclockwise rotation.

3.9.17 Container operations. The forklift with ROPS and FOPS shall be capable of entering, operating within, and exiting a 20-foot long x 8-foot wide x 8-foot high container conforming to ANSI MH5.1.1M, when the container is on the ground and when the container is supported on a chassis a minimum of 55 inches above ground. Access to the container-on-chassis shall be accomplished by traveling on a ramp conforming to MIL-R-52899.

3.9.18 Endurance. The truck shall be capable of completing 500 hours of performance testing without an endurance failure.

3.9.19 High speed. The forklift shall be capable of 4 hours of high speed operation when tested as specified in 4.6.2.32 without malfunction, damage, or permanent deformation, and without exceeding specified temperature limits for the cooling system (see 3.10.2.6), engine oil (see 3.10), transmission oil (see 3.11.1), universal joints (see 3.11) or hydraulic system (see 3.12).

3.10 Diesel engine. For the purpose of selecting the engine, the forklift truck shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, unless otherwise specified herein. All engine accessories shall be furnished by or approved by the engine manufacturer for this application. The engine shall be capable of operating on lubricating oils as specified in MIL-STD-1410. The engine furnished shall operate on fuel conforming to both VV-F-800, grade DF2 and MIL-T-83133. Unless otherwise specified herein, the engine shall meet all operational requirements and conditions specified herein without exceeding any of the operating limits specified in table I of MIL-STD-1410, or the engine manufacturer's published recommended operating limits. Emergency manual shutdown provisions shall be furnished when recommended by the engine manufacturer.

3.10.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the forklift under any of the operational requirements and conditions specified herein will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by the engine manufacturer.

3.10.2 Engine accessories.

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3.10.2.1 Starting priming system. Either a glow plug for each cylinder or an auxiliary fluid priming system operable from the operator's seat and conforming to MIL-E-52649, type III, shall be furnished. The primer actuator shall be a minimum of 1 foot away from the starter actuator. The size of the primer shall be in accordance with the engine manufacturer's recommendations. Primer actuator shall be electro-mechanically operated and provided with a temperature control in accordance with MIL-E-52649. Primer actuator shall be non-functional when engine is warm or running. The charged primer cylinder does not have to be furnished.

3.10.2.2 Alternator. A splashproof heavy-duty nominal 24-volt alternator in accordance with SAE J180 with reverse polarity protection and double lug mounting, shall be provided. The alternator shall be not less than 40-amp capacity and shall provide charge at engine low idle speed. The regulator shall be integral with the alternator.

3.10.2.3 Starting system. The starting system shall be 24 volts and shall include, as a minimum, a starting motor solenoid, a starter relay, a starting motor, necessary wiring and cable, and starting motor switch. A means shall be furnished to prevent energizing the starter motor whenever the engine is running. The forklift shall be provided with either an interlock in the starting system or other means to prevent energizing the starter motor except when the transmission is disengaged. The starter shall be in accordance with the engine manufacturer's recommendations. The starting motor and starter solenoid shall be splashproof.

3.10.2.4 Fuel system. The capacity of the fuel tank shall provide for not less than 10 hours operation when tested as specified in 4.6.2.34 and 4.6.2.35. The fuel tank and filler pipe shall be constructed of commercial quality, cold-rolled, long terne coated steel conforming to ASTM A 308. If the fuel line at any location is below the top of the fuel tank a hand operated fuel shutoff valve attached directly to the tank shall be provided. The valve shall be provided with double-ended arrows showing direction of valve operation, labeled at each end to indicate the functional result (i.e. open, close, etc.), the valve shall be labeled "FUEL SHUTOFF". The fuel tank filler cap shall be captive chained. The fuel fill port shall be labeled "Diesel Fuel" with cast or embossed letters unless a data plate in accordance with 3.32 is furnished near the fill pipe marked with "Diesel Fuel" in not less than 1 inch letter height. The tank shall accept fuel at the rate of not less than 17 gallons per minute and shall be located so that the tank can be filled from a 5-gallon can conforming to MIL-C-1283 by a person standing on the ground. The fuel tank shall be equipped with a drain plug at the lowest point of the tank. The drainplug shall be removable with hand tools without removal of any other component and shall be located such that the tank contents can be drained into a container. The fuel tank shall be protected by position or otherwise from external damage. Fuel lines shall provide for continuous operation at full throttle and maximum loads, including slope ascension. The fuel system shall include a fuel transfer pump, if required to maintain engine performance, water separator, fuel strainer, fuel filter, and a fuel injection pump, including provisions for priming. The water separator shall be located in the fuel line

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before the filters. The water separator shall have a drain valve which allows all of the water to drain from the separator when opened. A means shall be provided to allow the fuel system to be bled of all air after the separator has been drained, or the fuel filters changed.

3.10.2.5 Air cleaner. The engine shall be furnished with a heavy-duty, dry type, air cleaner in accordance with MIL-A-52363, class II or class III. The cleaner shall be of the duel element or two stage type and shall include a dust scavenger. The capacity of the air cleaner shall be in accordance with the engine manufacturer's recommendations. An air cleaner intake restriction indicator gauge shall be provided and shall be visible to the operator from the operator's seat.

3.10.2.6 Cooling system. When a liquid-cooled engine is furnished, the cooling system shall be of the closed pressure type, incorporating a radiator with a radiator pressure cap, hoses, thermostat, fan, and circulating pump. Under all operating conditions specified herein, the cooling system shall not boil when the ambient temperature is 120 °F or less. Drains shall be provided to completely empty the cooling system. Petcocks on all drains shall be accessible and contents should drain into a container. The cooling system fluid shall be a solution of ethylene glycol conforming to MIL-A-46153 and water. All fill ports shall be accessible by a person without any tools.

3.10.2.6.1 Fan. The fan shall discharge away from the operator. The fan shall rotate on antifriction bearings. The belts shall be clear of all other components within the adjustment range. Belts shall be accessible for easy replacement. Fan shall be protected by a welded wire or expanded metal fan guard in accordance with SAE J1308. If the fan operates below the 20-inch fording depth in 3.9.5, provisions shall be made for disconnecting the fan when necessary for fording operations without affecting the coolant circulation within the engine.

3.10.2.6.2 Water pump. When utilized, the water pump(s) shall be of the centrifugal type, shall be replaceable, and shall be equipped with shaft seals that will prevent leakage. The shaft shall be corrosion resistant when exposed to water and a mixture of ethylene-glycol and water. Either a lifetime lubrication system shall be provided, or accessible pressure lubrication fittings shall be provided.

3.10.2.6.3 Thermostat. When utilized, the thermostat shall be in accordance with the engine manufacturer's recommendations. The thermostat shall be removable.

3.10.2.6.4 Radiator. When utilized, the radiator assembly shall be a heavy duty type and shall consist of a core with an expansion tank and sediment tank. Where the top of the radiator is lower than the top outlet of the engine, a surge tank may be used. The radiator cap shall meet the requirements of SAE J164 and be accessible without removal of the engine hood. A sign warning "CAUTION - DO NOT OPEN WHEN HOT" shall be placed near the radiator cap. The radiator shall be mounted to prevent its being damaged due to shock and racking experienced under any operation of the forklift specified herein. The radiator shall be protected

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by a grilled guard or by the counterweight. A shield shall be provided, if necessary, to protect the radiator from mud and gravel thrown by the tires. The cooling system shall include baffles and shroud to prevent recirculation of cooling air.

3.10.2.6.5 Gaskets, seals, and packing. When utilized, gaskets, seals, and packing used in the cooling system shall be made of materials capable of resisting deterioration from mixtures of water and ethylene glycol base antifreeze conforming to MIL-A-46153.

3.10.2.6.6 Engine coolant hose. When utilized, all engine coolant hoses shall be SAE J20, heavy-duty type. Hoses shall clear the forklift structure and other components under all operating conditions specified herein. All hose clamps shall be of the double wrap stainless steel type in accordance with SAE J536, type F, style 1, 2, or 4.

3.10.2.7 Governor. A governor shall be provided to limit the engine speed to not more than the intermittent duty speed recommended by the engine manufacturer when the forklift is operated under any condition specified herein.

3.10.2.8 Exhaust system. The exhaust system shall be protected against entry of rain. The back pressure of the exhaust system, including muffler, shall in no case exceed the maximum recommended by the engine manufacturer when measured 2 inches beyond the header outlet at any load up to rated net continuous load. The exhaust system shall be arranged or shielded to prevent burns to the operator and maintenance personnel while working on the step surfaces of the forklift. Exhaust discharges shall be directed into the discharge air from the fan to minimize fumes at the operator's station. The exhaust system shall have a spark arresting efficiency of not less than 80 percent when tested in accordance with Forest Service Standard 5100-1 or SAE J350. When a spark arrestor is provided it shall be provided with a means for cleaning out trapped particles.

3.10.2.9 Crankcase ventilation system. The crankcase ventilation system shall prevent water from entering the engine during fording operations. When the crankcase vapors are directed into the induction system, means shall be provided on turbocharged engines to prevent choking of turbocharger blades. No vapors or condensation shall be deposited on or come in contact with the air-cleaner element.

3.10.2.10 Oil filter. Oil filter(s) of the heavy-duty full flow type or heavy-duty full flow and partial flow type shall be furnished in accordance with the engine manufacturer's recommendations. A label in accordance with MIL-STD-1472 specifying filter type shall be placed near the point of filter installation. Oil lines, when applicable, to and from the filters shall be installed to minimize vibration and shall be of flexible hose meeting the requirements in SAE J30 of 30R2 hose type 2, with reusable fittings.

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3.10.2.11 Tachometer drive. When the Diagnostic Connector Assembly (DCA) is not provided (see 3.42), a heavy-duty key type tachometer drive shall be provided and shall have a 5/32-inch female connection in accordance with SAE J678 for diagnostic purposes. The speed of the tachometer drive shall be 1/2 of the crankshaft speed.

3.10.2.12 Oil sampling valves. Oil sampling valves conforming to MIL-V-81940 shall be provided on the engine, transmission, and hydraulic system. The valves shall be located in such a way as to insure that personnel will not be exposed to danger and a true representation of oil that is flowing is obtained when taking oil samples with the engine running. The sampling valves shall be labeled adjacent to the valve in accordance with MIL-STD-1472 indicating the type of oil sampled.

3.11 Drive assembly. The drivetrain assembly shall consist of all components necessary to transmit power from the engine to all four wheels. All rotating shafts and axles shall be supported on antifriction bearings. All gears shall operate in lubricant, and a standard pressure grease lubrication system shall be provided for other friction parts. Gear housing shall be equipped with filler plugs and magnetic drainplugs. Differential ring and pinion gears shall be adjustable. Drive shafts shall be provided with a minimum of two heavy-duty universal joints and a nonbinding slip joint. Drivetrain components shall conform to and shall be installed in accordance with each component manufacturer's recommendations. Drivetrain universal joints shall not exceed 180 °F when the forklift is tested as specified herein.

3.11.1 Transmission. The forklift shall be provided with either a hydrodynamic or hydrostatic transmission as defined in SAE J645. The transmission shall be protected from damage when the operator moves the directional control to the opposite direction while the forklift is in motion. The forklift shall be capable of providing a smooth directional change when the directional control lever is moved to the opposite direction in any of the provided gear ranges. The transmission oil shall conform to MIL-L-2104, grade 10, or MIL-L-46167. The fill tube shall be functionally labeled "transmission fill" in accordance with MIL-STD-1472. All transmission component material shall be compatible with these oils. The transmission fluid shall be filtered by filter(s) having replaceable elements in accordance with the transmission manufacturer's recommendations. The transmission shall be provided with a disconnect control. The disconnect control shall disengage the transmission while allowing the engine to be operated throughout the entire speed range.

3.11.1.1 Hydrodynamic transmission. The hydrodynamic transmission shall provide controlled application of the clutches to provide smooth shifting, limiting slippage which would damage the clutches. A minimum of 3 forward and 2 reverse gears shall be provided. All transmission gears shall be constant mesh. Oil pressure for actuation of the transmission clutches, for torque converter supply, and for transmission lubrication shall be supplied by a mechanically driven pump. The hydrodynamic transmission shall allow the forklift to be accelerated from a standstill to maximum speed by starting in the lowest gear and

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shifting through consecutive gears to the highest gear all while under continuous full engine throttle with continuous flow of torque to the driving wheels. This acceleration shall be accomplished on level ground and without damaging the transmission. The transmission shall be capable of withstanding maximum stall operation for a minimum of 30 seconds without damage or permanent deformation and without exceeding the fluid temperature limit of 250 °F with an initial fluid temperature of not less than 157 °F in an ambient temperature of not lower than 25 °F.

3.11.1.2 Hydrostatic transmission. The hydrostatic transmission shall be a closed loop type with a reversible variable displacement pump and a reversible variable or reversible fixed displacement motor. A charge pump shall be provided and shall be gear or shaft driven by the engine and shall provide sufficient oil under pressure to retard cavitation of the pump and motor and to maintain a constant, smooth control lever response independent of operating conditions performed at full throttle engine speeds. With an oil temperature of 140 ±5 °F and new filter, the vacuum at the charge pump inlet port shall not exceed 5 inches Hg when tested in accordance with 4.6.2.21.1. A shuttle valve or equivalent means shall be provided to automatically prevent loss of high pressure oil from the closed loop pump/motor circuit during the transition of rapid change in direction of the forklift. A set of reduction gears with a minimum of two speed ranges capable of being shifted on the move under full engine torque may be used in conjunction with the hydrostatic transmission. When a gear range selector is provided it shall be accessible to the operator in the normal operating position and shall be labeled to indicate speed ranges. The transmission shall allow the forklift to be accelerated from standstill to maximum speed under continuous full engine throttle with continuous flow of torque to the driving wheels on level ground and shall be accomplished without damaging the transmission. Oil temperatures in the transmission shall not exceed 180 °F when measured at the pump case drain and motor case drain under any operating conditions specified herein. When a heat exchanger is required to meet temperature limits stated herein, means shall be provided to insure recommended case pressures of the pump and motor are not exceeded during all operating conditions specified herein including cold weather start-up. The transmission shall be capable of withstanding maximum stall operation for a minimum of 30 seconds without damage or permanent deformation and without exceeding the fluid temperature limit of 180 °F. The system shall withstand all operations specified herein without permanent deformation, damage, or leakage.

3.11.2 Rear axle oscillation. The rear axle shall oscillate freely such that the total vertical travel of each rear wheel shall be a minimum of 12 inches (six inches above and six inches below the horizontal axle centerline).

3.11.3 Traction control. The front differential shall be of the driver controlled locking type to insure power is transmitted to the wheel(s) having traction when the opposite wheel(s) loses traction or is off the ground. The

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differential shall automatically disengage after release of the driver control. An engaged warning light labeled "front differential is locked" (in accordance with MIL-STD-1472) shall be installed near the warning light and in a position clearly visible to the driver.

3.12 Hydraulic system. The hydraulic system shall consist of all fluid power components necessary for the operations of the forklift, including the steering system and hydrostatic transmission system but does not necessarily include the brakes or the hydrodynamic type power train. All fill openings shall be labeled "Hydraulic Fill" in accordance with MIL-STD-1472. The working hydraulic system (lift, tilt, fork functions and steering) maximum operating pressure (relief valve setting) shall not exceed 3000 psi and if furnished the hydrostatic transmission relief valve setting shall not exceed 6100 psi. The system fluid shall conform to MIL-L-2104, grade 10 or MIL-L-46167. All hydraulic system component material shall be compatible with these oils. The temperature of the hydraulic fluid shall not exceed 100 °F above the ambient during all conditions from -25 to +120 °F while conducting operations and tests specified herein. Hydraulic fittings on pumps, valves and cylinders shall be either the O-ring, straight thread type, as specified in SAE J514 or the 4-bolt flange type as specified in SAE J518 for 1 inch and smaller, and only the 4-bolt flange type as specified in SAE J518 for ports larger than 1 inch. The hydraulic system shall be flushed free of foreign matter including weld slag and spatter. The connection ports of the hydraulic components shall be sealed by the component manufacturer. The hydraulic system shall permit the rated load to be lowered at the rate of speed specified in 3.9.8 in the event of failure of or damage to the hydraulic line supplying the cylinder. A straight thread O-ring boss diagnostic port conforming to SAE J514 shall be provided at the pump(s) outlet or pressure inlet of each control valve.

3.12.1 Hydraulic lines. Hydraulic lines shall consist of tubing, flexible hose and steel fittings. Hose or a combination of hose and tubing shall be used between components that are not mounted to a nonflexing, rigidly connected structural member or rigidly connected subassembly. All lines shall be routed to provide the minimum number of bends, ease of maintenance, maximum protection and provision shall be made to prevent damage of lines due to chafing. The hose installation, such as angle of fitting, and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines, other than suction lines, and cylinder bleeder lines, shall be considered pressure lines.

3.12.1.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517 or MIL-H-52471. Permanently attached type pressure hose couplings are acceptable as original equipment hose assemblies. The truck shall be compatible with hose assemblies with field attachable reusable fittings in place of each hose assembly with permanently attached couplings when tested as specified herein. All fittings shall be 37-degree flare, female swivel or 4-bolt split flange as specified in SAE J516. Hose ID shall be limited to 1/4, 3/8, 1/2, 3/4, 1, 1-1/4,

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1-1/2, and 2 inches. The hose selected shall have a rated operating pressure that is greater than the hydraulic circuit maximum relief valve setting. The hose installation shall not cause the hose to be bent beyond the minimum bend radius specified in SAE J517.

3.12.1.2 Pressure tubing. Pressure tubing shall be in accordance with SAE J524 or SAE J525 and shall meet the requirements specified herein. Pressure tube sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, and 2 inches. Tubing shall have a minimum burst pressure that is equal to or greater than 3.0 times the system maximum working pressure. Tubing shall be securely clamped. All bends shall be smooth without flattening, kinking or wrinkling of the tube. The term "without flattening" is defined as follows:

> Throughout the bend, the tube maximum diameter shall be equal to or greater than the tube nominal diameter, and the tube maximum diameter minus the tube minimum diameter shall be not greater than 15 percent of the nominal tube diameter.

3.12.1.3 Pressure tube fittings. Unless otherwise specified herein pressure tube fittings shall be 37-degree flared and straight thread fittings conforming to SAE J514 or 4-bolt split-flange type conforming to SAE J518. Jump size tees may be used. The 4-bolt split-flange type connections may terminate in either a flange head that incorporates a SAE J518 plain mating face or a flange head that integrally incorporates four boltholes and an O-ring groove for mating to a SAE J518 port face. Pressure tube fittings that incorporate the SAE J518 connection shall contain a boss into which the tube shall be brazed or welded. Pressure tube fitting sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, and 2 inch. Pressure tube fittings shall have a minimum burst pressure 3 times the maximum operating pressure.

3.12.1.4 Suction lines. A pump inlet suction line shall include provision for flexibility when there is vibration or movement between the reservoir and the pump. The suction line shall be sized so that the pump inlet pressure does not drop below atmospheric pressure by more than 5 inches of mercury (vacuum) at 140 ±10 °F oil temperature and at governed engine speed. The inlet suction line installation shall include no stress concentration at the end fittings and no kinks. If hose is used it shall conform to SAE 100R4 hose found in SAE J517. Suction hose clamps shall be of the double wrap stainless steel type in accordance with SAE J536, type F, style 4, or saddle worm thread type with stainless steel inner liner, or a heavy duty T-bolt clamp. Hose fittings in accordance with 3.12.1.1 may be used on the suction hose.

3.12.1.5 Return lines. Return lines shall be in accordance with SAE J517.

3.12.2 Hydraulic pumps. The hydraulic pump(s) shall have a 1000 hour Omega rating which is less than the system filter Beta Ten rating.

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3.12.3. System filtration. System filter(s) shall be furnished either in the pressure line, return line (suction line of charge pump allowed for filtration of hydrostatics), or combinations (as specified in 3.12.3.1 or 3.12.3.2). Labels in accordance with MIL-STD-1472 identifying each filter shall be provided near the points of filter installation.

3.12.3.1 Return line filters. The filter shall maintain a full flow condition for the duration of reliability testing as specified in 4.6.2.35 and endurance testing specified in 4.6.2.34. The filter bypass may open for system oil at a temperature less than 70 °F. All return oil shall pass through the filter except relief oil which may return directly to the reservoir. The filter shall be located so that the element may be removed without the removal of other components for examination or replacement without spilling or draining oil from the reservoir. At the maximum system oil flow through the filter, including lowering rated load at engine idle; and an oil temperature of 150 ±5 °F, the pressure drop across a new filter assembly shall not exceed 10 psi. An automatic bypass shall be provided in the system filter and shall open at not less than 10 psi greater than the pressure drop measured across the new filter assembly and it shall permit full flow at a pressure drop not greater than 35 psi. The filter housing shall have a minimum proof pressure of 100 psi without permanent deformation, damage, or leakage. The system filter shall be equipped with a contamination indicator which may be remotely mounted. The indicator shall be clearly readable without the removal of any components. As a minimum, the indicator shall show when the filter element is clean and when the system filter is beginning to bypass, requiring a change of the element.

3.12.3.2 Pressure filter. The pressure filter(s) shall conform to 3.12.3.1 except that proof pressure test shall be 150 percent of the maximum system operating pressure.

3.12.4 Reservoir. A hydraulic reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be mounted to prevent vibrations that could result in leaks, loosening of fasteners, brinelling of material, or cracks. The location shall not impair operator visibility or access to either side of the operator's compartment. The reservoir shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended and sufficient free air capacity to prevent oil being discharged through the reservoir air vent when maximum return flow of oil is surged into the reservoir from the system. The Add and Full marks on the sightglass(es) shall indicate the correct oil level when the truck and forks are on level ground, and the oil is at normal operating temperature. The fillercap and tube shall be located to provide for filling from a standard 5 gallon container conforming to MIL-C-1283 by a person standing on the ground or a surface on the forklift intended for standing, and to prevent entrance of contaminants throughout filling and checking operations. The reservoir filler neck shall contain a removable 50 mesh, or finer strainer retained in place. The filter tube shall be not less than 1-inch inside diameter. The reservoir pump inlet line suction openings except for the auxiliary steering pump opening, shall be located a

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minimum of 2 inches above the bottom of the reservoir. When furnished, the auxiliary steering pump inlet line suction opening shall be located a minimum of 1/2 inch above the bottom of the reservoir. Except for sealed reservoir, a pressurized breather with a 30 psi maximum relief valve and 1-inch of mercury (vacuum) breaker shall be furnished. The breather shall have a nominal rating of 40 microns. The breather shall preclude the entrance of water. The reservoir shall provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. All return line flow except the vent line shall discharge into the reservoir at a level which is below the oil level in the reservoir. When filters are located within the reservoir, a removable 50 mesh, in-tank suction strainer with bypass valve shall be installed in the suction line of the reservoir. The reservoir shall have suction flanges and one or more access holes (see 3.7) with covers for inspection, cleaning and maintenance of the entire inside.

3.12.5 Relief valve. A relief valve shall be provided to limit the pump pressure to the maximum required to perform the necessary system functions. Pressure equal to the relief valve setting shall be generated only when a cylinder is bottomed or when the system is subjected to shock loading.

3.12.6 Hydraulic control valves. The control valve spools shall be spring centered with sufficient force to return and retain the levers at the center position throughout all forklift operations. The spool connection eye end shall not bind while operating as specified herein. The exposed control-valve spool surfaces shall show no evidence of corrosion or pitting when tested as specified herein. The contractor shall present information during first article tests, that the control valves were tested in accordance with procedures specified in SAE J747 and J1117. The inlet to outlet port pressure differential shall not exceed 3 percent of the maximum operating pressure at rated flow.

3.12.7 Cylinders. All hydraulic system and steering cylinders shall be fabricated from either seamless steel tubing or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. Cylinders shall be so located as to provide for ease of maintenance and replacement. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device to prevent foreign material and fluids from entering around the piston rod and damaging the seal and other cylinder components. All hydraulic system and steering cylinders shall perform as specified herein without buckling or bending. Each piston shall have a bearing surface concentric with the piston to carry the piston side loads against the cylinder walls. The bearing surface shall be of a material that will not scar, score, or gall the inside surface of the cylinder. All cylinder rods shall be hard chrome plated. All cylinders shall withstand a pressure that is equal to 1.5 times the system relief valve setting without evidence of rupture, permanent deformation, damage, internal or external leakage.

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3.12.8 Contamination levels for hydraulic system. When tested as specified in 4.6.2.21.3, the particulate matter in the hydraulic fluid removed from the system shall not exceed the ISO code level of 19/16 in accordance with SAE J1165.

3.12.9 External leakage. After completing tests specified in 4.6.2, the degree of external hydraulic system leakage shall not exceed a class 3, for dust-free conditions and a class 3D for dusty conditions of SAE J1176.

3.12.10 Hose reels. When hydraulic hose reels are used on the forklift to maintain hose tension during mast extension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges.

3.13. Mast and fork carrier assembly. Mast uprights shall be furnished and shall be of the telescopic, roller type. The lift shall be hydraulically actuated, and the lift cylinder(s) shall be mounted and positioned so that it does not protrude forward of the plane formed by the front faces of the uprights. The mast assembly shall provide the operator with maximum visibility. Chains shall be furnished and shall be installed in such a manner that they cannot be displaced from the roller until they have been raised a minimum of one-half the chain side plate height. Chains used in the lifting mechanism shall have a factor of safety of not less than 4.6 to 1 based on average ultimate strength. Adequate means shall be provided in the uprights to compensate by replacement or adjustment for wear which may develop between lateral rolling contacts. Carriers shall be equipped with adjustable thrust bearings or side rollers to provide for forces imposed by off-center loading. The forks and fork carrier shall be positioned on the lift mechanism in such a manner that, the fork carrier and all loads wider than the forklift and positioned at the rear of the forks, will not come in contact with the front tires or any other portion of the forklift in any possible fork position. The fork carrier shall not exceed 40 inches in height. The fork carrier width shall not exceed the width of the forklift when the fork carrier is centered.

3.13.1 Sideshift. The lift mechanism shall be equipped with a hydraulically powered sideshift mechanism that shall provide lateral displacement of the rated load from a centered position to either side, not less than (0.5W-18) inches, where "W" equals the forklift width. The sideshift function shall provide for full sideshift travel specified with the forks positioned at any point within the extreme of the fork spread specified. When the forks are sideshifted to either maximum sideshift position, the truck width shall not be increased by more than 1 inch. There shall be no protruding bolts or appendages beyond the side plane of the sideshift carriage assembly. Time to sideshift rated load horizontally from one extreme to the other extreme shall not exceed 15 seconds at engine low idle and shall not exceed 6 seconds at engine high idle. The sideshift mechanism shall have sufficient capacity to sideshift rated load uphill at maximum oscillation position of the carriage. No permanent deformation or damage shall result when the sideshift mechanism is operated a minimum of 100 cycles when tested in accordance with 4.6.2.23.

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3.13.2 Load backrest. Two metal backrests shall be provided. One backrest, removable for container operations without disturbing any other component, shall be provided on the forklift carriage. In conjunction with fork and carriage design, it shall provide a vertical rear guard of at least 48 inches high, measured from the load carrying surface of the forks to the top of the backrest and shall have a width equal to the width of the carriage. Storage provision on the forklift shall be provided for the removable backrest. The removable backrest shall not exceed 37 lbs in accordance with MIL-STD-1472. Another backrest shall provide a vertical rear guard, for container operations, at least 20 inches high measured from the load carrying surface of the forks to the top of the backrest and shall not unduly obstruct the operator's visibility and the spacing between vertical members shall be no more than 6 inches. The backrest(s) may be eliminated if fork and carriage assembly will provide protection equal to the above.

3.14 Forks. The forklift shall withstand, without permanent deformation or damage, 300 percent of the rated load on the forks at a 24-inch load center, with the fork carrier centered; and 200 percent of the rated load on the forks at a 24-inch load center at full shift to either side when tested in accordance with 4.6.2.15. Forks shall be attached to the fork carrier in a manner which prevents lateral movement of the forks, except when fork position is intentionally changed by the operator. Fork positioning shall be performed by one person without using tools or shall be performed hydraulically. The fork carrier shall permit the forks to be moved together to an inside-to-inside dimension of not more than 4-inches (between the inside edges of the forks) and moved apart as far as an outside-to-outside dimension of not less than 34 inches (over the outside edges of the forks). The fork length shall be 40 inches +0, -1/2 inch. The fork width shall be not more than 5 inches. The fork thickness shall be not more than 1.5 inches at the heel tapered to not more than 0.5 inch at the tip. The taper shall be on the underside of the forks and shall be not less than 14 inches long. The fork tips shall be chamfered or rounded to break corners and provide smooth edges.

3.15 Wheels and tires.

3.15.1 Wheels. The forklift shall be equipped with four demountable wheels. The wheels shall be fastened to the hubs in accordance with the axle manufacturer's recommendations for off-road applications. Wheel rims shall be in accordance with the TRA Yearbook. Split rims shall not be used.

3.15.2 Tires. All tires shall be new and unused, wide base, off-the-road type, and of identical size and construction. The tires shall be pneumatic, low pressure, tubeless, furnished with traction-type tread or radial tires with traction-type tread. Tire loadings shall be in accordance with applicable tables of the TRA Yearbook, or loadings in accordance with published literature by the rim or wheel manufacturer and the tire manufacturer at the 5 mph rating. Valves and caps shall be in accordance with the TRA Yearbook. Large bore valves shall be furnished. A valve adapter shall be furnished to reduce the large bore to the standard passenger vehicle size bore. Valve caps shall be metal with elastomer innerseal. Hydroinflated tires are not acceptable.

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3.16 Wheel guards. Wheel guards shall be provided for all four wheels to prevent foreign objects and road splash being discharged in the direction of the operator. Each guard shall support a weight of 300 pounds without permanent deformation or damage.

3.17 Steering system. The steering method shall be either the hinged-frame (articulated) type or Ackermann type. The steering shall be actuated by hydraulic cylinders. The steering system shall permit control of the forklift while traveling at any speed up to and including the maximum specified travel speed (see 3.9.4). Ackermann type steering shall be provided with an adjustable, mechanical followup mechanism to maintain a positive ratio between the amount the steering wheel is turned and the amount the forklift turns, except that a followup need not be provided for a hydrostatic steering system. The power steering actuating device shall be located so as to absorb all road shock without transmitting the shock effect to the steering wheel. The forklift shall operate without loss of operator control when tested as specified in 4.6.2.12. For forklifts with hinged-frame steering, a steering framelock in accordance with SAE J276 shall be provided except that the framelock shall be painted in accordance with 3.33.

3.17.1 Steering control. The steering system shall be controlled by a steering wheel. The effort required to turn the steering wheel shall not exceed 14 pounds pull at the rim. The steering wheel shall require not more than 4-1/2 turns to move the wheel from one stop to the opposite stop. The steering wheel shall require not more than 5 seconds for the maximum range of movement at low idle engine speed and shall require not less than 1.5 seconds at governed engine speed. There shall be a minimum of 7 inches measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point of the unoccupied seat cushion. There shall be a minimum of 3 inches between the rim of the steering wheel and the control levers for all positions of the levers. The steering wheel shall be capable of withstanding a minimum 300 pound force applied perpendicular to the plane of the steering wheel, and a minimum 150 pound force applied tangentially to the rim of the steering wheel, without permanent deformation or damage. The steering system shall be operable with a dead engine when the forklift is towed. When a four wheel Ackermann steering system is furnished, a selector shall be provided to select four wheel and front wheel only steering modes.

3.17.2 Emergency steering. Emergency steering shall be required only if the truck cannot be steered to the curb under control of the operator with a dead engine in accordance with 4.6.2.12.3. The force to turn the steering wheel shall not exceed 50 pounds at the rim in the event of power assist failure. If required, the emergency steering conforming to SAE J53 shall be provided for retaining steering control in the event of engine failure or steering power source failure when traveling at any speed to the maximum speed.

3.18 Brake system. Split system service and parking brakes shall be provided and shall be independently actuated systems. If a separate hydraulic system for the brakes is utilized, the brake system shall be filled with a fluid that is listed on the qualified product list of MIL-B-46176. Brake linings shall be non-

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asbestos. If air brakes are provided, air brake service and emergency line coupling conforming to SAE J318 shall be provided on the rear of the vehicle; a relay emergency valve shall also be provided. An alcohol evaporator shall be provided in the air system if air brakes are furnished.

3.18.1 Service brakes. Unless otherwise specified herein, foot controlled disk service brakes shall be provided for all wheels and shall conform to SAE J1152 for loaders. The brakes shall be capable of bringing the forklift, with rated load on the forks, to a complete stop without any wheel leaving the ground from a speed of 15 mph as follows:

a. Dry - within 28 feet with fishtailing of no more than 1 foot when tested as specified in 4.6.2.18.
b. Fade - within 28 feet with fishtailing of no more than 1 foot when tested as specified in 4.6.2.18.
c. Wet - within 38 feet with fishtailing of no more than 1 foot when tested as specified in 4.6.2.18.

Pedal force while making any stop shall not exceed 70 pounds. The brake system shall provide for bleeding of air from hydraulic lines and for draining of moisture from the compressed air system, as applicable. The brake system shall withstand an operator's brake application force of not less than 250 pounds at the pedal, a minimum of 10 times, without permanent deformation or damage. Automatic brake adjustment which maintains correct lining to disk adjustment without looseness or lockup shall be provided on each brake.

3.18.2 Parking brake(s). Parking brake(s) shall hold, but not necessarily stop the forklift without rated load on the forks when the forklift is stopped on a 30 percent grade in both the ascending and descending position. The parking brake(s) shall be hand operated and equipped with a locking device. The manual control shall be located within reach of the operator and in a noninterfering position. Hand lever pull to set the parking brake shall not exceed 30 pounds. The parking brake shall be an independent mechanical friction brake mounted on the drive shaft or axle, or a separate isolated system actuating the service brakes. The parking brake(s) shall be effective when the transmission is disengaged for towing (see 3.1.3.1). Adjustment of parking brake shoes or bands shall be accomplished without the removal of any assembly. An indicator light which warns the operator when the parking brake is applied, shall be furnished in a position clearly visible to the operator and shall be labeled in accordance with MIL-STD-1472.

3.18.3 Emergency braking provisions. The emergency braking provisions shall provide that any single failure (other than a common friction means such as brake linings) in the service brake system shall not leave the forklift without braking capability. For test purposes the forklift shall provide not less than six consecutive "dead engine" stops with rated load while descending a 30 percent grade. The emergency braking provisions shall be provided through the service brake system. Means shall be provided to release the brakes for towing.

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3.19 Operator access. Operator access systems to the forklift shall be in accordance with minimum requirements of SAE J185. Access and egress to the operator's compartment shall not require the operator to climb any higher than 6 inches above the floor of the operator's compartment.

3.20 Operator's seat. A cushion seat and backrest shall be furnished in accordance with the minimum requirements specified in SAE J899 and located in accordance with SAE J1163 to permit 90 percent of U.S. male/female operator population (SAE J833) to reach controls in the optimum areas of the forklift as defined in SAE J898 by operator clothed in NBC and cold-wet gear. The seat shall be easily adjustable horizontally and vertically (not to exceed 3 inches) by the operator without use of hand tools so the operator can see the tip ends of the forks when attempting to pick up pallets. The seat cushion and backrest shall be foam rubber or polyurethane, covered with slip covered vinyl upholstery having a minimum weight of 40 ounces per yard for 54-inch width.

3.21 Hood. The engine shall be protected by a hood cover and sidepanels. Either the hood cover or sidepanels shall be quick opening and shall provide complete accessibility to all engine accessories without requiring the removal of any additional forklift components. The hood cover, if lift up type, shall be held in the open position in a safe manner by counterweight, spring-and-linkage mechanism, latch, or by a brace which cannot be inadvertently released while servicing the forklift. The quick opening hood cover or sidepanels shall be capable of being opened or removed by one person and shall not exceed 37 pounds in weight. Elastomer hood latch or paddle latch(s) shall be provided on the quick opening hood cover or sidepanels.

3.22 Instrument panel. The instrument panel shall be located so that the operator can read the instruments from his normal operation position. The instrument panel shall be installed on an incline at a minimum of 5 degrees to a maximum of 80 degrees. On this panel shall be mounted the charging system monitor, engine oil-pressure-monitor, engine-coolant temperature monitor, torque converter or transmission-fluid temperature monitor, fuel gauge, and any other instruments used, unless otherwise specified herein. Universal symbols shall be furnished in accordance with SAE J1362. Instruments shall conform to SAE J209 and be weather-tight. All gauges shall be flush mounted and shall be provided with switch-controlled illumination. A lateral level indicator for the truck shall be provided on the instrument panel.

3.22.1 Charging system monitor. A remote indicating ammeter gauge shall be furnished. The ammeter shall be of sufficient capacity to indicate variations in current supply or demand experienced by the forklift, exclusive of starter requirements. The gauge shall be marked in red in the critical zone and green in the normal zone and shall be labeled with appropriate universal symbology.

3.22.2 Engine oil pressure monitors. A remote indicating engine oil pressure gauge which is marked in red in the low pressure zone and a low pressure red indicator light shall be furnished. Means shall be provided to check operation of the light.

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3.22.3 Engine coolant temperature monitor. For liquid cooled engines, a remote indicating engine coolant temperature gauge marked in red in the high temperature zone shall be furnished.

3.22.4 Transmission temperature monitor. A remote indicating torque converter or transmission temperature gauge or a high temperature red indicator light shall be furnished. If an indicator light is furnished, means shall be provided to check operation of the light.

3.22.5 Hourmeter. An operational hourmeter which registers the number of engine operating hours up to a minimum of 9999 hours shall be provided. The hourmeter shall be accessible for reading without moving any component of the truck.

3.22.6 Fuel gauge. An electric, thermally stabilized, remote indicating fuel gauge shall be provided.

3.22.7 Air pressure monitor. When a compressed air system is furnished, an air pressure gauge shall be furnished and an audible alarm shall indicate when air pressure is inadequate for proper brake operations.

3.23 Controls. Each control shall be provided with symbolic labels conforming to SAE J1362 and labeled as a function and direction of movement either on plates conforming to 3.32 or embossed or depressed on or near the control. Labeling for levers which control any part of the lift mechanism shall conform to ANSI MH11.3. The labels shall be visible to the operator from his normal operating position. All transmission, hydraulic, engine, and service brake controls shall be located in the operator's compartment and shall be within the zone of comfort specified in SAE J898. All other controls for the forklift shall be located in the operator's compartment and shall be within the zone of reach specified in SAE J898. Unless otherwise specified herein, the controls shall conform to and have the recommended clearances of MIL-STD-1472. Electric controls shall be weather-tight. All hand operated controls shall be operable by persons wearing cold-wet gloves. Mechanical push-pull remote controls, when furnished, shall be of waterproof, heat resistant, antifriction type, conforming to requirements for marine cables specified in SAE J917, except as specified herein. As a minimum, all exposed parts of the control cables shall be fabricated of corrosion-resistant steel that has been passivated as per QQ-P-35. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding. Bend radii shall be as recommended by the cable manufacturer. The "as-installed" mechanical efficiency of each installation shall be not less than 70 percent and the no-load friction shall not exceed 5 ounces per foot. The control manufacturer's recommended load ratings for continuous operation shall not be exceeded for any required operation of the control specified herein. Only controls associated with the cab, such as windshield wiper and heater controls, may be mounted on the cab structure.

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3.23.1 Light switches. "On-off" light switches shall be furnished on the instrument panel. One switch shall control the forward floodlamps and one switch shall control the rear floodlamps as specified in 3.26.2. One light switch conforming to MS51113 and clearly labeled as to its specific function (i.e., front headlights, rear floodlamps, etc.) shall be provided for the floodlamps (see 3.26.2), taillights (see 3.26.2.2), blackout headlight (see 3.26.2.1), two blackout stoplight-taillights (see 3.26.2.1), and instrument panel lights (see 3.22). All internal and external lights, the horn, and backup alarm shall be made inoperable when blackout lights are used, except for warning lights, such as temperature and pressure lights, which shall remain operative in the blackout mode.

3.23.2 Engine run control. An engine run control shall be furnished and shall be lever type and waterproof. It shall have not less than two positions: "on and off". The switch shall control operation of monitors, and electrical accessories, except horn.

3.23.3 Staring system control. The starting system solenoids (see 3.10.2.3), shall be operated by a pushbutton starter switch. The starter shall be inoperable when the engine run control is in the "off" position.

3.23.4 Braking and declutch pedal(s). The pedal(s) shall be located for convenient left and right foot operation and shall be a combination declutch and braking pedal(s). The forklift shall be capable of achieving specified lifting, lowering, tilting, and sideshift rates with a rated load while the brake pedal is depressed and the transmission control is positioned both in the forward and reverse positions, without moving the truck. The effort required to activate the declutch control shall be not less than 25 pounds. When one pedal is furnished it shall be a minimum of 8 inches wide and shall have an all around clearance of not less than 2 inches. When two pedals are furnished each pedal shall be a minimum of 3 inches wide and shall have an all around clearance of not less than 2 inches. Grease fittings or permanent lubrication shall be provided for lubricating moving parts of all pedals and shafts.

3.23.5 Accelerator control. The accelerator control shall be installed for comfortable right foot operation and shall be located to the right of, and shall be free of interference from the brake pedal. The accelerator shall have an all around clearance of not less than 3 inches.

3.23.6 Transmission controls. Selective forward and reverse and speed range controls shall be provided and shall be located for convenient left hand operation. Forward, reverse, and speed ranges shall be labeled in accordance with SAE J1362. Movement of the control lever in a forward or upward direction shall cause forward motion of the forklift. Transmission control levers and linkage shall withstand, without permanent deformation or damage, a force of 100 pounds at the handgrip and the shock and vibration encountered during operation specified herein. The force necessary for the movement of the hand control levers shall not exceed 20 pounds and shall have a minimum force of 2 pounds at the handgrip.

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3.23.7 Hydraulic controls. Lift, tilt, oscillation, and sideshift, hydraulic controls shall be self centering lever(s), located for convenient right hand operation. All control levers and linkage shall withstand, without permanent deformation or damage, a force of 100 pounds at the handgrip and the shock and vibration encountered during operations specified herein. The force necessary to move any control to the end of its travel shall not exceed 20 pounds and shall have a minimum force of 2 pounds at the handgrip. Total control lever travel from one end to the other shall not exceed 14 inches, measured at the center of the handgrip. Clearance between control handgrips shall be not less than 2 inches in any control position. When separate lift and tilt levers are used, the tilt control shall be at the right of or below the lift control, as viewed by the operator. Control functions and predominant control motion shall be as follows (function listed, followed by control motion - the first applicable motion listed, depending on lever mounting, shall apply):

a. Lift load - toward the operator, rearward, or upward (opposite direction to lower).
b. Tilt load rearward - toward the operator, rearward, or in the same rotational direction as fork movement (opposite direction to tilt forward).
c. Sideshift to right - to the operator's right, rearward, or upward (opposite direction to sideshift left).
d. Oscillate clockwise (as viewed by operator) - to the operator's right, rearward, or upward (opposite direction to oscillate counterclockwise).

3.24 Battery and battery mounting.

3.24.1 Batteries. A minimum of two 12 volt batteries shall be furnished. Batteries shall be in accordance with MS52149, type "6TL" and shall be readily accessible for service. The batteries shall be shipped charged and dry. Electrolyte shall be furnished in a separate container as specified in MIL-STD-162. The batteries shall be negative grounded in accordance with SAE J538. The battery terminals shall be provided with guards to prevent accidental shorting.

3.24.2 Battery mounting. Batteries shall be located so they can be cleaned, serviced, and removed without removing any component except the quick-release battery box cover if a battery box is furnished. Battery supports, holddown, and areas around the installation which could possibly be affected by dripping or seepage of acids shall be protected with a coating conforming to TT-C-494, type II. The battery shall be mounted in such a manner as not to interfere with access to engine components (accessories). Battery mounting shall provide for complete support over the entire base of the battery and shall be in such a position that the level of the electrolyte is directly visible without removing the battery from its mounting bracket or requiring the use of tools. Battery restraining clamps shall not be designed for or dependent upon special battery shapes and shall be provided to hold the battery in a fixed position. The battery compartment shall have provision for drainage, and shall have provision for gas venting at or near the top of the compartment. No-sag cover and positioning shall be protected

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against short circuiting. The cables shall be protected by rubber grommets or insulated passages at entry to the battery box. A pictorial battery diagram shall be provided in a location which is readily visible when batteries are being serviced.

3.24.3 Battery cables. Battery cables shall be sized in accordance with SAE J541 for heavy duty use. Positive and negative cables shall be color-coded and labeled; positive colored red and labeled "+" and negative colored black and labeled "-". Terminals conforming to MS 75004 shall be provided on the ends of the cables connected to the battery.

3.24.4 Slaving components. The truck shall be equipped with a 24-volt slave receptacle conforming to MS52131. The slave receptacle shall permit charging of the truck batteries and slave starting of the engine from an external power source and shall also provide a power source for charging and slaving other equipment. The slave receptacle shall be installed on the exterior of the truck near the battery enclosure and shall be accessible to personnel standing on the ground. The voltage potential of the slave receptacle shall be identified on a plate furnished adjacent to the receptacle which reads "24-volt slave receptacle". The plate shall be in accordance with 3.32.

3.25 Pintle hook. A rear mounted pintle hook conforming to MS51335 shall be provided. The pintle hook shall be located on the vertical centerline of the forklift not less than 31 inches nor more than 37 inches above ground level. The pintle hook mounting shall be of sufficient strength to withstand, without permanent deformation or damage, the drawbar pull induced when the pintle hook is restraining forward motion at torque converter stall (see 3.11.1) and when used in its normal operations.

3.26 Electrical components and circuits. All wiring shall be protected by position or other means to prevent the wires from being damaged due to rough terrain operations.

3.26.1 Circuit breaker. Each electrical circuit shall be protected with a circuit breaker in accordance with SAE J553. Fuses are not acceptable.

3.26.2 Lights. The forklift shall be provided with not less than two floodlamps, and two headlamps for forward illumination and not less than two floodlamps for rearward illumination. The lights shall be shock mounted in elastomer ring housings. The floodlamps shall be Trade No. 4578 in accordance with SAE J598. The headlamps shall be in accordance with SAE J598. All forward floodlamps shall be positioned such that they are visible by a direct line of sight from an observer positioned 20 feet forward of the truck when rated capacity load (see 6.6.2) is in carry position (see 6.6.3). The two forward headlamps shall be mounted on the fenders and the two forward floodlamps shall be mounted on the ROPS near the front of the truck. The floodlamps mounted on the ROPS shall be

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provided with stud and terminal connectors near the place of attachment of the ROPS to the truck chassis. All floodlamps and headlamps shall be adjustable in accordance with SAE J598. All lights shall be positioned within the outline of the truck.

3.26.2.1 Blackout lighting. A separate wiring harness may be provided for the blackout lights. One blackout headlight conforming to MS51318 shall be mounted on the extreme left within the plan outline of the forklift at the front, positioned to provide illumination with minimum obstruction when the forks are in the carry position (see 6.6.3). The blackout headlight shall be adjustable in accordance with SAE J598 for headlamps. Two blackout stoplight-taillights conforming to MS51330 or MS52125 shall be mounted adjacent to the rear taillights, and shall be mounted in 6-inch-diameter holes or shall be provided with guards. Each blackout stoplight-taillight lens face shall be recessed not less than 1/2 inch behind the hole or face of the guard. Two blackout marker lights conforming to MS52126 shall be mounted on the front, one on each side of the vertical centerline of the truck, at the same level, and as far apart as practicable.

3.26.2.2 Taillights. Two taillights-stoplight assemblies in accordance with SAE J585 and SAE J586 shall be installed, one on the left rear and one on the right rear of the forklift. Each lens face shall be recessed not less than 1/2 inch back of a protecting member.

3.26.3 Interior lighting. The truck shall be equipped with gauge lighting or indicators which are readily visible to the full range of user personnel and shall be adequately lighted for night operation. Interior lights, gauges and instruments, to include warning lights, shall not emit energy out of the 380-700 nanometer wavelength range. All warning indicators shall be light emitting diodes.

3.26.4 Horn. The forklift shall be equipped with a horn. The horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant to prevent entry of moisture when operated or stored outdoors under all weather conditions. The horn button shall be mounted on the steering wheel.

3.27 Personnel cab. When specified (see 6.2), the forklift shall be equipped with an insulated, weathertight personnel cab, which may be combined with ROPS and FOPS specified in 3.5.1. The cab clearances shall be in accordance with SAE J154. The cab shall withstand all operational shock and vibration without deformation or damage under the operating conditions specified herein. The cab shall be constructed of metal and shall be equipped with a minimum of one door and another emergency exit (either door or window that can be opened from both the inside and outside) on the other side of the cab. All doors shall have nonlocking closing latches and opening stops. The doors shall either slide and latch at each end of travel or shall swing open a minimum of 90 degrees. All hardware shall be of the heavy duty truck or industrial type, resistant to corrosion. Front and rear

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window defrosters, operating in conjunction with the heater blower (see 3.27.4), or not less than two ventilation fans, not less than 200 cfm inclosed in a wire guard, shall be furnished. Fans shall be directionally adjustable to direct air flow toward all glazing. As a minimum, each fan shall have an "on-off" switch.

3.27.1 Safety glazing. As a minimum, safety glazing shall include a windshield, door windows and a rear window. Other windows, as necessary, shall be provided to give an upright seated operator all around visibility and visibility of the fork carrier in all possible fork positions. When a roof window is provided, it shall be sloped to provide for water drainage and guarded as necessary to meet the FOPS requirements (see 3.5.1). Safety glazing shall conform to ANSI Z26.1, type AS1 for the windshield and roof window and type AS1 or AS2 for other windows. Fixed or hinged safety glazing shall be sealed in waterproof, stress absorbent molding and movable windows shall be sealed on extra-heavy-type, pile-lined channels.

3.27.2 Interior. The interior of the cab shall be fitted with insulation retarding both heat loss and noise intrusion. Heavy-duty floor mats shall be furnished. All cab openings shall be sealed with elastomer grommets, seals, weather stripping, or body compound. Elastomer components shall display a rating of two or better conforming to ASTM D 1171 for ozone chamber testing.

3.27.3 Windshield and window wipers. The cab shall be equipped with at least two powered windshield assemblies to clear the windshield and the rear window in accordance with the performance requirements of SAE J198. Wiper arms and blades shall be of the heavy duty commercial type. The wiper arms shall provide tension on the glazed surface of at least 1 ounce for each inch of wiper arm length. Air motors shall be plumbed to exhaust air outside of the cab. The operator controls shall be installed within the zone of reach as defined in SAE J898 within the cab.

3.27.4 Heater. A recirculating air type heater shall be furnished. The heater shall have a rating of at least 19,000 Btu/hour. The heater controls shall be within the zone of reach as defined in SAE J898 and shall provide for variable selection of blower speeds. If a fuel-fired heater is furnished, it shall be in accordance with SAE J1024.

3.28 Electromagnetic interference. The electromagnetic interference emission characteristics of the forklift shall meet the limits specified in MIL-STD-461, part 8, class C1, group II, except that a 20 dB relaxation is permitted in the emission limits. The operators horn is exempt from electromagnetic interference requirements.

3.29 Lubricants. Unless otherwise specified herein, the forklift shall be serviced with military lubricants selected by the contractor in accordance with table I, SAE J754. Run-in and testing shall be performed with these lubricants as applicable for the ambient operating temperature. A lubrication data plate identifying military lubricants and applicable temperature ranges shall be provided (see 3.32) and mounted on each forklift.

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3.29.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected accessible position. Fittings shall be accessible by all grease guns conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand operated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with 3.7.

3.29.2 Enclosures with integral reservoir of lubricant. Enclosures such as the gear cases, transmission housing and engine crankcase, which contain a reservoir of liquid lubricants for the lubrication of the parts enclosed, shall be equipped with dipsticks marked in accordance with SAE J614, check plugs not less than 1/2-inch pipe size, or sight gauges to determine the acceptable level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drains, except planetaries, shall be fitted with a magnetic drainplug having either tapered pipe threads or shoulder and gasket. The drainplug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage, except from planetary hubs, shall be into a suitable container on the ground when the forklift is on level ground. Integral tubes or troughs may be used to convey the lubricant from the drain to the container. Accessibility to the drainplug, the filling means, and the lubricant level checking device shall be provided without the removal or adjustment of accessories or parts. Plates equipped with hand operated fasteners may be removed.

3.30 Fungus and moisture resistance. Electrical circuity, including all components and connections except as specified below, shall be protected from the effect of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition specified in 3.30.1.

a. Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter that would interfere with the adherence or function of the varnish.

3.30.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.31 Slinging and tiedown provisions. Permanently affixed slinging and tiedown provisions that enable the forklift to be lifted in its normal travel position and to be fastened to the floor or deck of a transportation medium shall be provided.

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Provisions shall conform to MIL-STD-209, classes 1 and 2 or class 3 for type II equipment, including the air transportation and helicopter external transport requirements.

3.32 Identification and instruction plates. Each forklift shall be identified in accordance with MIL-STD-130 on identification plates conforming to MIL-P-514, type I, style 1, composition C (GG-P-455, type I, grade A, class 1) material. Colors to be black with natural characters except for hazard and warning plates and labels. The identification plate shall also be marked "CARC MONTH/YEAR" with the month and year the vehicle was painted with CARC paint. A shipping data plate and instruction plates/labels shall conform to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material. The shipping data plate shall indicate the silhouette of the forklift in the transport position indicating the center of gravity and the location and capacity of the slinging and the tiedown provisions. The instruction plates/labels shall include warnings, cautions, and diagrams describing any special or important procedures to be followed in operating, towing, or servicing the forklift. Instruction plates and labels shall conform to the labeling paragraph of MIL-STD-1472. The plates shall be furnished and mounted by screws, drive screws, bolts, or rivets in conspicuous protected locations by the contractor.

3.33 Treatment and painting. The portions of the forklift normally painted, both internally and externally, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F for ferrous metals and type G for non-ferrous metals and fiberglass top coat color camouflage green 383. Primer in accordance with MIL-P-53030 may be used in lieu of MIL-P-52192. Finish coat in accordance with MIL-C-53039 (polyurethane) may be used in lieu of MIL-C-46168. Vendor parts and components received by the prime contractor with either alkyd primer or alkyd paint coating may be coated with above specified primers and paints applied directly over existing alkyd primer or paint without additional surface preparation except for cleaning, if the base alkyd primer or paint meets applicable military federal specifications and if the base alkyd primer or paint is allowed to cure for a period of 14 days. This exception for vendor parts and components is not intended to supersede the requirement that the complete truck be primed and painted in accordance with MIL-T-704 using the primers and paints specified above. If another color is specified (see 6.2), the above system shall be overcoated with MIL-C-46168 or MIL-C-53039 of the desired color. All surfaces of the forklift which exceed 400 degrees F during operation shall not be primed or painted as specified above, but shall be painted in the appropriate color of the base coat, or camouflage pattern scheme if applicable, with paint conforming to MIL-P-14105.

3.34 Markings. Registration and other markings shall be applied in accordance with MIL-STD-642. The gross weight of the forklift shall be stenciled on each side of the forklift. The TRA Yearbook prescribed tire pressure shall be stenciled "TP (specify) LBS" in a suitable location adjacent to each tire. All suitable slinging and tiedown points (see 3.31) shall be marked either "Lifting Point", "Tiedown Point" or "Lift and Tiedown". Unless otherwise specified (see 6.2), vehicle markings and their locations for Air Force use shall be in

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accordance with MIL-STD-1223. Registration numbers will be furnished by the contracting officer. The markings shall be lusterless paint conforming to MIL-C-53039 or MIL-C-46168. Black, FED-STD-595, color chip No. 37030, shall be used on a green or brown background. Green 383, color chip No. 34094 shall be used over a black background. Markings shall be block or stencil type characters not more than 1 inch and not less than 3/4 inch in height.

3.35 Toolbox. A toolbox shall be fabricated integrally inside the cab area or a separate toolbox made of steel having a minimum nominal thickness of 0.048 inch, shall be provided. The tool box shall not allow entry of water when tested in accordance with 4.6.2.28. The toolbox shall have a hinged lid and a hasp accommodating a padlock shackle of 3/8 inch diameter, accessible and operable by personnel wearing gloves. The toolbox shall be bolted or welded to the forklift in a protected position, shall have provisions for drainage of liquids and shall accommodate any maintenance tools furnished. The toolbox shall be treated and painted in accordance with 3.33. When specified (see 6.2), the tool box shall accommodate an NBC protective ensemble (see 3.41.1).

3.36 Decontamination bracket assembly. A bracket assembly MS53052 shall be furnished and mounted. The location of the bracket shall be such that it shall not interfere with normal operation of the truck and it is accessible to personnel and provides clearance to remove and emplace a 5-gallon container conforming to MIL-C-1283 having a maximum weight of 54 lb. The bracket shall not be installed in close proximity to exhaust or heat sources.

3.37 Workmanship.

3.37.1 Castings and forgings. All parts, components, and assemblies of the forklift which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.37.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly, except in the performance of designed heat treatment. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.37.3 Welding.

3.37.3.1 Welder qualifications. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by one of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

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ASME	Boiler and Pressure Vessels Code, Section IX, Welding and Brazing Qualifications.
AWS D1.1	Structural Welding Code - Steel, Section 5, Qualification.
AWS D14.3	Specification for Welding Earthmoving and Construction Equipment, Section 5.
MIL-STD-248	Welding and Brazing Procedure and Performance Qualification.

Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. The test results shall be made available for review by the contracting officer or the contracting officer's representative and shall be maintained for a period of four years after contract completion. The Government reserves the right to require the retesting of any welder or welding operator at its discretion, if a problem with weld quality or retesting data exists.

3.37.3.2 Welding procedures. All welding shall be performed in accordance with written procedures as defined by one of the following codes:

ASME	Boiler and Pressure Vessels Code.
AWS D1.1	Structural Welding Code - Steel.
AWS D14.3	Specification for Welding Earthmoving and Construction Equipment.
MIL-STD-248	Welding and Brazing Procedure and Performance Qualification.

All welds shall be inspected and tested in accordance with the criteria for workmanship and welding quality as stated in one of the above codes. Copies of written welding procedures shall be available at the workstation for operator reference and inspection.

3.37.4 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers, lockwashers, or other positive locking devices shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.37.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492 or IFI Standard No. 114 for Break Mandrel Blind Rivets.

3.37.6 Machine work. Tolerance and gauge for metal fits shall conform to the limits specified herein and to the standards of the Materials Handling Equipment Industry or the Construction Equipment Industry.

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3.38 Nuclear-biological-chemical (NBC) protection. Unless otherwise specified (see 6.2), the forklift shall be operable by personnel wearing the following complete protective ensemble:

a. Chemical-biological mask, MIL-M-51282.
b. Chemical-biological hood, MIL-H-51291.
c. Chemical protective suit, MIL-S-43926.
d. Chemical protective glove set, MIL-G-43976.
e. Chemical protective footwear, MIL-F-43987.

The operator shall be capable of actuating all controls (i.e., levers, switches, pedals, knobs, and handles) without interference caused by the ensemble that is detrimental to the operation of the forklift. The forklift shall complete one cycle in 1.5 hours or less on the container/trailer loading course as specified in 4.6.2.40. The protective ensemble, as listed above, will be furnished as Government furnished equipment (GFE) in order to permit performance of the tests required under test method no. 35.

3.39 M16 mounting. Unless otherwise specified (see 6.2), a mounting bracket, for securing an M16 rifle, conforming to TACOM Drawing number 12276986 shall be provided and mounted in a suitable location in or near the operators compartment.

3.40 Vehicle weight classification sign kit. Unless otherwise specified (see 6.2), the contractor shall apply the Government furnished vehicle weight classification numbers to each vehicle utilizing a vehicle weight classification number kit conforming to MIL-S-40626. The kit shall be located on the front of the vehicle in a location approved by the contracting officer. The contracting officer shall provide the classification numbers to be displayed on the vehicle.

3.41 Government-furnished equipment (GFE). Unless otherwise specified (see 6.2), the following item shall be furnished by the Government (see 6.5).

Vehicle weight classification sign kit, NSN 9905-00-565-6267, for each truck.

3.41.1 Government-loaned equipment. Unless otherwise specified (see 6.2), the following items shall be loaned by the Government (see 6.5.1):

a. NBC protective ensemble for test method No. 35.

1. Chemical-biological mask, NSN 4240-00-926-4200.
2. Chemical-biological hood, NSN 4240-00-999-0420.
3. Chemical protective suit, NSN 8415-00-407-1062.
4. Chemical protective glove set, NSN 8415-01-033-3519.
5. Chemical protective footwear, NSN 8430-01-021-5978.

b. STE/ICE-R for test in 4.6.2.41.1.

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c. Cold-wet gloves.

1. Shell, NSN 8415-00-269-5700.
2. Liner, NSN 8415-00-682-6575.

d. Ramp, MIL-R-52899 for use in tests 4.6.2.22, 4.6.2.34, 4.6.2.35, 4.6.2.40.

3.42 Diagnostic connector assembly (DCA) measurement capabilities. Unless otherwise specified (see 6.2), the forklift shall incorporate an easily accessible DCA located in or near the operator's compartment for interface with the Simplified Test Equipment for Internal Combustion Engines Reprogrammable (STE/ICE-R) test equipment. The DCA shall be in accordance with TACOM drawing No. 12258941 and shall be labeled "Diagnostic Connector Assembly" in accordance with MIL-STD-1472. All requirements for DCA shall be in accordance with the STE/ICE-R design guide for vehicle Diagnostic Connector Assemblies, report No. CR-82-588-003 revision 1. The DCA shall have the capabilities for measuring the minimum functions defined in table 2-2 of report No. CR-82-588-003 revision 1. Additional test parameters to be monitored in either the DCA or Transducer Kit (TK) mode shall be as follows: fuel return pressure, engine oil pressure, engine oil filter pressure drop, starter solenoid voltage, and transmission oil pressure. Test points identified to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. Those additional test parameters which are inaccessible for monitoring without removal of other components in the TK mode shall be monitored in the DCA mode. Fittings/adapters required for interface with the STE/ICE-R test set when monitoring in the TK mode, shall be permanently installed on the forklift to insure that adapters in the STE/ICE-R test set readily mate with the TK mode test point locations. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. A fuel shut-off method shall be provided for conducting compression unbalance checks. The contractor shall also provide STE/ICE-R vehicle test cards in the format identified in TACOM Drawing 12258955 addressing both DCA and TK measurements, and shall incorporate the test cards into the forklift technical manuals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The

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absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3 and 4.4).
b. Quality conformance inspection (see 4.5).
c. Inspection comparison (see 4.7).
d. Inspection of packaging (see 4.8).

4.2.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.2.1.

4.3 First article inspection. Prior to examination and test of the forklift specified in 4.3.1 and 4.3.2, the following shall be performed:

a. Service the forklift with the oils, greases, and referee diesel fuel conforming to MIL-F-46162, as specified herein and designated for use in the ambient temperatures at which the tests will be conducted.
b. Conduct the run-in as prescribed (see 4.6.2.37).
c. Set the hydraulic system's relief pressure as specified by the contractor and in 3.12 and record. Any change made to this setting shall constitute failure of all tests.
d. Furnish a schedule of maintenance to be followed during all testing of the forklifts. A list of tools required to perform this maintenance shall also be provided.

4.3.1 First article examination.

4.3.1.1 Pretest examination. Prior to testing, all first article forklift(s) shall be examined for the defects marked "X" in column 1 of table II. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test disassembly and examinations. Upon successful completion of all tests specified in 4.3.2, the truck that underwent endurance testing (see 4.6.2.34) shall be subjected to the tests marked "X" in column 2 of table III and 4.6.2.39. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. Upon successful completion of examinations specified in 4.3.1.1, one first article forklift shall be subjected to the endurance, reliability, and post-test disassembly tests (see 4.6.2.34, 4.6.2.35, and 4.6.2.39). The second first article forklift shall be subjected to endurance and reliability tests (see 4.6.2.34 and 4.6.2.35). The third first article forklift shall be subjected to the tests marked "X" in column 1 of table III except for the

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endurance, reliability, and post-test disassembly tests (see 4.6.2.34, 4.6.2.35,and 4.6.2.39). The DCA interface test (see 4.6.2.41.1) shall be conducted as the first test for the third first article forklift. Following completion of all previously specified tests for the third first article forklift, drain and flush the entire fuel system and refill the fuel system with fuel conforming to MIL-T-83133 and subject the forklift to the travel speed, longitudinal gradeability, 500 hours of reliability, and paragraphs (a) and (b) of the post-test disassembly tests (see 4.6.2.6, 4.6.2.10, 4.6.2.35, 4.6.2.39). Except as otherwise specified herein, test conditions shall apply throughout testing. Failure of any test shall be cause for rejection. Failure of a first article forklift to meet any requirement specified herein shall be cause for rejection of the first article forklifts.

4.4 Inspection failure. Failure of a first article or production truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3 or 4.5 shall be cause for rejection of the first article truck(s), and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that all trucks accepted prior to the completion of first article inspection are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each forklift shall be examined for the defects marked "X" in column 2 of table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each forklift shall be tested as specified in column 3 of table III. Failure of any test shall be cause for rejection.

4.5.3 Sampling.

4.5.3.1 Unit of product. A unit of product is defined as a fully-serviced forklift.

4.5.2.2 Lot. A lot shall consist of not more than 25 units of product. Lots shall be formulated from forklifts which have successfully completed the individual examination and tests as specified in 4.5.1 and 4.5.2.

4.5.3.3 Sample tests. One sample shall be selected from each lot and shall be subjected to the tests specified in 4.6.2.4, 4.6.2.5, 4.6.2.6, 4.6.2.9, 4.6.2.17, 4.6.2.18 and 4.6.2.23. Any sample failing any test shall be cause for rejection of the lot.

4.6 Inspection schedule.

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4.6.1 Examination. Forklifts shall be examined as specified herein in accordance with table II. An "X" in the column indicates the examinations that shall be conducted.

TABLE II. Examination schedule.

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	X	101. Parts or components missing or do not function.	Section 3
X	X	102. Assembly incorrect or not complete.	Section 3
X	X	103. Towbar or safety chains missing or not as specified.	3.1.3.1
X	X	104. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the forklift trucks.	3.3
X	-	105. Dissimilar metals not insulated as specified.	3.3.1
X	-	106. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.	3.3.1.1
X	X	107. Contractor does not have documentation available for identification of material, material finishes or treatments.	3.3.1.1
X	X	108. Drain holes were not provided.	3.3.1.2
X	-	109. Screw threads not as specified.	3.3.2
X	X	110. Tire material not as specified.	3.3.3
X	-	111. Hose material not as specified.	3.3.4
X	X	112. Safety provisions not as specified.	3.5
X	X	113. ROPS and FOPS missing or not as specified.	3.5.1
X	X	114. Seatbelt missing or not as specified.	3.5.2
X	X	115. Backup alarm missing or not as	3.5.3
X	X	116. Fire extinguisher and bracket not as specified.	3.5.4
X	X	117. Mirrors not as specified.	3.5.5
X	X	118. Human factors not as specified.	3.6

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TABLE II. Examination schedule. (Cont'd)

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	-	119. Maintainability provisions not as specified.	3.7
X	-	120. Calculated flotation index not as specified.	3.9.2
X	X	121. Engine not as specified.	3.10
X	X	122. Starting priming system missing or not as specified.	3.10.2.1
X	X	123. Alternator not as specified.	3.10.2.2
X	X	124. Starting system not as specified.	3.10.2.3
X	-	125. Fuel system not as specified.	3.10.2.4
X	X	126. Air cleaner and restriction indicator not as specified.	3.10.2.5
X	-	127. Cooling system not as specified.	3.10.2.6
X	X	128. Fan not as specified.	3.10.2.6.1
X	-	129. When utilized, water pump not as specified.	3.10.2.6.2
X	-	130. When utilized, thermostat not as specified.	3.10.2.6.3
X	X	131. Radiator not as specified.	3.10.2.6.4
X	-	132. Gaskets, seals, and packing not as specified.	3.10.2.6.5
X	-	133. Engine coolant hose not as specified.	3.10.2.6.6
X	-	134. Governor not as specified or not set to manufacturer's recommended speed.	3.10.2.7
X	-	135. Exhaust system not as specified.	3.10.2.8
X	-	136. Crankcase ventilation system not as specified.	3.10.2.9
X	X	137. Oil filters not as specified.	3.10.2.10
X	-	138. Tachometer drive not as specified, when furnished.	3.10.2.11
X	X	139. Oil sampling valve and label not	3.10.2.12
X	-	140. Drive assembly not as specified.	3.11
X	X	141. Transmission not as specified.	3.11.1
X	-	142. Traction control not as specified.	3.11.3
X	-	143. Hydraulic system not as specified.	3.12
X	-	144. Hydraulic lines not as specified.	3.12.1

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TABLE II. Examination schedule. (Cont'd)

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	-	145. Pressure hose assemblies not as specified.	3.12.1.1
X	-	146. Pressure tubing not as specified.	3.12.1.2
X	-	147. Pressure tube fittings not as specified.	3.12.1.3
X	-	148. Suction line not as specified.	3.12.1.4
X	-	149. Return lines not as specified.	3.12.1.5
X	-	150. System filtration not as specified.	3.12.3 thru 3.12.3.2
X	-	151. Reservoir not as specified.	3.12.4
X	-	152. Relief valve not as specified.	3.12.5
X	-	153. Hydraulic control valves not as specified.	3.12.6
X	-	154. Cylinders not as specified.	3.12.7
X	X	155. External leakage exceeds limits specified.	3.12.9
X	X	156. Hose reels not as specified.	3.12.10
X	X	157. Mast and fork carrier not as specified.	3.13
X	X	158. Backrests not as specified.	3.13.2
X	X	159. Forks not as specified	3.14
X	X	160. Wheels not as specified.	3.15.1
X	X	161. Tires not as specified.	3.15.2
X	X	162. Wheelguards not as specified.	3.16
X	-	163. Steering and steering control not as specified.	3.17
X	X	164. Brake system and brake fluid not as specified.	3.18
X	-	165. Service brakes not as specified.	3.18.1
X	-	166. Parking brakes not as specified.	3.18.2
X	-	167. Emergency braking provisions not as specified.	3.18.3
X	X	168. Operator access not as specified.	3.19
X	X	169. Seat not as specified.	3.20
X	X	170. Hood not as specified.	3.21
X	X	171. Instrument panel not as specified.	3.22
X	X	172. Charging system monitor missing or not as specified.	3.22.1
X	X	173. Engine oil pressure monitors missing or not as specified.	3.22.2

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TABLE II. Examination schedule. (Cont'd)

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	X	174. Engine coolant temperature monitor missing or not as specified.	3.22.3
X	X	175. Transmission temperature monitor missing or not as specified.	3.22.4
X	X	176. Hourmeter missing or not as specified.	3.22.5
X	X	177. Fuel gauge missing or not as specified.	3.22.6
X	X	178. Air pressure monitor missing or not as specified.	3.22.7
X	X	179. Controls not as specified.	3.23
X	X	180. Light switches missing or not as specified.	3.23.1
X	X	181. Engine run control not as specified.	3.23.2
X	X	182. Starter system control not as specified.	3.23.3
X	-	183. Braking and declutch controls not as specified.	3.23.4
X	X	184. Accelerator control not as specified.	3.23.5
X	X	185. Transmission controls not as specified.	3.23.6
X	X	186. Hydraulic controls not as specified.	3.23.7
X	X	187. Batteries not as specified.	3.24.1
X	X	188. Battery mounting not as specified.	3.24.2
X	-	189. Battery cables not as specified.	3.24.3
X	X	190. Slave receptacle missing or not as specified.	3.24.4
X	X	191. Pintle hook missing or not as specified.	3.25
X	-	192. Circuit breakers missing or not as specified.	3.26.1
X	X	193. Lights missing or not as specified.	3.26.2
X	X	194. Blackout lighting missing or not as specified.	3.26.2.1

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TABLE II. Examination schedule. (Cont'd)

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	X	195. Taillights missing or not as specified.	3.26.2.2
X	X	196. Interior lighting not as specified.	3.26.3
X	X	197. Horn missing or not as specified.	3.26.4
X	X	198. Personnel cab not as specified.	3.27
X	-	199. Glazing not as specified.	3.27.1
X	X	101a. Cab interior not as specified.	3.27.2
X	X	102a. Windshield and window wipers not as specified.	3.27.3
X	X	103a. Heater missing or not as specified.	3.27.4
X	X	104a. Lubrication not as specified.	3.29
X	-	105a. Lubricants fittings not as specified.	3.29.1
X	X	106a. Filling, draining, and checking provisions for lubricant enclosures missing or not as specified.	3.29.2
X	-	107a. Fungus and moisture resistance provisions not as specified.	3.30, 3.30.1
X	X	108a. Slinging and tiedown provisions (and frame lock if applicable) missing or not as specified.	3.31
X	X	109a. Identification marking missing or not as specified.	3.32
X	X	110a. Instruction plates missing or not as specified.	3.32
X	X	111a. Treatment and painting not as specified.	3.33
X	X	112a. Marking not as specified.	3.34
X	X	113a. Toolbox not as specified.	3.35
X	X	114a. Decontamination bracket assembly not as specified.	3.36
X	X	115a. Castings and forgings not as specified.	3.37.1
X	X	116a. Metal fabrication not as specified.	3.37.2
X	X	117a. Welding not as specified.	3.37.3 thru 3.37.3.2

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TABLE II. Examination schedule. (Cont'd)

First article	Quality conformance	Defect	Requirement paragraph
1	2	3	4
X	X	118a. Welding certification not as specified.	3.37.3.1
X	X	119a. Bolted connections not as specified.	3.37.4
X	X	120a. Riveted connections not as specified.	3.37.5
X	X	121a. Machine work not as specified.	3.37.6
X	X	122a. M16 mounting bracket not as specified.	3.39
X	X	123a. Vehicle weight classification sign kit not as specified.	3.40

4.6.2 Tests. Test shall be conducted in accordance with table III. An "X" in the column indicates the tests that shall be conducted.

4.6.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Only that maintenance established by the contractor and submitted as maintenance schedule prior to commencement of the tests shall be performed during the tests. All tests shall be conducted with ROPS and FOPS installed. Safety and human factor considerations (see 3.5 and 3.6), shall be evaluated throughout testing. Test methods referenced in tests specified herein are those contained in the appendix. Operation of the forklift shall be with fuels, oils, lubricants, and antifreeze specified herein as applicable for the temperature conditions existing at place of test. All tests shall be conducted at the contractor's recommended tire pressure.

TABLE III. Test schedule.

First article	Post-test	Quality conform-ance	Test	Test Paragraph	Requirement paragraph
1	2	3	4	5	6
X	X	X	Test conditions.	4.6.2.1	3.5, 3.6, 3.29
X	-	X	Engine.	4.6.2.2	3.10

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TABLE III. Test schedule. (Cont'd)

First article	Post-test	Quality conformance	Test	Test Paragraph	Requirement paragraph
1	2	3	4	5	6
X	-	-	Weight and dimensions.	4.6.2.3	3.1.2.1 thru 3.1.2.4, 3.9.11 thru 3.9.14
X	X	-	Lifting speed and lowering.	4.6.2.4	3.9.7, 3.9.8
X	-	-	Curb clearance circle.	4.6.2.5	3.9.9
X	X	-	Travel speed.	4.6.2.6	3.9.4, 3.10.2.8, 3.11.1
X	-	-	Underclearance, angles of approach and departure.	4.6.2.7	3.1.2.5
X	-	-	Upright tilt, tilt speed, tilt cavitation, mast clearance.	4.6.2.8	3.13, 3.9.10
X	X	-	Drift.	4.6.2.9	3.9.15
X	-	-	Longitudinal gradeability.	4.6.2.10	3.9.3
X	-	-	Dynamic stability, parking brake, emergency brake.	4.6.2.11	3.9.1.2, 3.18.2, 3.18.3
X	-	-	Static stability.	4.6.2.11.1	3.9.1.1
			Steering.	4.6.2.12	-
X	-	-	Obstacle course No. 1.	4.6.2.12.1	3.17
X	-	-	Obstacle course No. 2.	4.6.2.12.2	3.17
X	-	-	Emergency steering.	4.6.2.12.3	3.17.2
X	-	-	Steering control.	4.6.2.12.4	3.17.1
X	-	-	Highway & rail transportability.	4.6.2.13	3.1.3.1, 3.1.3.4
X	-	-	ROPS and FOPS.	4.6.2.14	3.5.1
X	-	-	Overload.	4.6.2.15	3.14
X	-	-	Torque converter.	4.6.2.16	3.11.1.1, 3.25
X	-	-	Transmission disconnect.	4.6.2.17	3.11, 3.11.1, 3.23.4, 3.10.2.6, 3.10
X	-	-	Service brake.	4.6.2.18	3.18.1
X	-	-	Maintenance.	4.6.2.19	3.7
X		-	Air transportability.	4.6.2.20	3.1.3.2, 3.1.3.3
			Hydraulic system.	4.6.2.21	-

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TABLE III. Test schedule. (Cont'd)

First article	Post-test	Quality conformance	Test	Test Paragraph	Requirement paragraph
1	2	3	4	5	6
X	-	-	Pump inlet pressure.	4.6.2.21.1	3.12.1.4
X	-	-	Hydraulic filter and filter assembly.	4.6.2.21.2	3.12.3, 3.12.3.1, 3.12.3.2
X	X	X	Hydraulic system cleanleness.	4.6.2.21.3	3.12.8
X	-	-	Hydraulic cylinders.	4.6.2.21.4	3.12.7
X	-	-	Pump contaminant tolerance.	4.6.2.21.5	3.12.2
X	-	-	Hydraulic fittings.	4.6.2.21.6	3.12.1.1
X	X	X	External leakage	4.6.2.21.7	3.12.9
X	-	-	Container operations.	4.6.2.22	3.6, 3.9.17
X	-	-	Sideshift, oscillation, and fork spread.	4.6.2.23	3.13.1, 3.9.16, 3.14
X	-	-	Electromagnetic interference.	4.6.2.24	3.28
X	-	-	Slinging provisions.	4.6.2.25	3.31
X	-	-	Tiedown provisions.	4.6.2.26	3.31
X		X	Noise level measurement.	4.6.2.27	3.6.1
X	-	-	Rain.	4.6.2.28	3.10.2.8, 3.4.4
X	-	-	Wheel guards.	4.6.2.29	3.16
X	-	-	Low temperature (-25 °F).	4.6.2.30	3.4.2, 3.4.3, 3.10, 3.10.2.6, 3.12
X	-	-	Fording.	4.6.2.31	3.9.5, 3.10.2.3 3.10.2.9
X	-	-	High speed.	4.6.2.32	3.9.19
X	-	-	Controls.	4.6.2.33	3.23.6, 3.23.7
X	-	-	Endurance.	4.6.2.34	3.9.18, 3.10.2.4
X	-	-	Reliability	4.6.2.35	3.8, 3.9.6, 3.10, 3.10.2.4, 3.10.2.6, 3.11, 3.12

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TABLE III. Test schedule. (Cont'd)

First article	Post-test	Quality conformance	Test	Test Paragraph	Requirement paragraph
1	2	3	4	5	6
X	-	-	Maintainability.	4.6.2.36	3.7
X	-	X	Run-in.	4.6.2.37	Section 3
X	-	-	Rear axle oscillation.	4.6.2.38	3.11.2
X	X	-	Post-test.	4.6.2.39	Section 3
X	-	-	Nuclear-Biological-Chemical.	4.6.2.40	3.38
X	-	X	Diagnostic connector assembly capability.	4.6.2.41	3.42
X	X	X	DCA - STE/ICE-R interface.	4.6.2.41.1	3.42

4.6.2.2 Engine. The diesel engine shall be tested or certified in accordance with production engine tests and production control tests of MIL-STD-1410.

4.6.2.3 Weight and dimensions. Determine the weight and dimensions in accordance with test method No. 1. Nonconformance to 3.1.2.1, 3.1.2.2, 3.1.2.3, 3.1.2.4, 3.9.11, 3.9.12, 3.9.13, or 3.9.14, shall constitute failure of this test.

4.6.2.4 Lifting speed and lowering speed. Test the forklift in accordance with test method No. 2. Nonconformance to 3.9.7 or 3.9.8 shall constitute failure of this test.

4.6.2.5 Curb clearance circle. Determine the curb clearance circle in forward and reverse as defined in Field Test Procedure of SAE J695 with rated load in carry position. Nonconformance to 3.9.9 shall constitute failure of this test.

4.6.2.6 Travel speed. Test the forklift in accordance with test method No. 3. Speeds not as specified in 3.9.4 shall constitute failure of this test. Inability to shift as specified in 3.11.1.1 or 3.11.1.2 as applicable shall constitute failure of this test. Exhaust back pressure not in accordance with 3.10.2.8 shall constitute failure of this test.

4.6.2.7 Underclearance, angles of approach, and departure. Test the forklift in accordance with test method No. 4. Nonconformance to 3.1.2.5 shall constitute failure of this test.

4.6.2.8 Upright tilt, tilt speed, tilt cavitation, mast clearance. Test the forklift in accordance with test method No. 5. Nonconformance to 3.9.10 or 3.13 shall constitute failure of this test.

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4.6.2.9 Drift. Test the forklift in accordance with test method No. 6. Nonconformance to 3.9.15 shall constitute failure of this test.

4.6.2.10 Longitudinal gradeability. Test the forklift in accordance with test method No. 7. Inability of the forklift to ascend the slope as specified in 3.9.3 shall constitute failure of this test. Exhaust back pressure not in accordance with 3.10.2.8 shall constitute failure of this test.

4.6.2.11 Dynamic stability, parking brake, emergency brake. Test the forklift in accordance with test method No. 8. Nonconformance to 3.9.1.2, 3.18.2, or 3.18.3 shall constitute failure of this test.

4.6.2.11.1 Static stability. Test the forklift in accordance with ANSI B56.6. Nonconformance to 3.9.1.1 shall constitute failure of this test.

4.6.2.12 Steering.

4.6.2.12.1 Obstacle course No. 1. Test the forklift in accordance with test method No. 9, 3a and 3b. Inability of the forklift to negotiate the test course or the hitting of any two markers for any run shall constitute loss of operator control and shall constitute failure of this test.

4.6.2.12.2 Obstacle course No. 2. Test the forklift in accordance with test method No. 9, 3c. Inability of the forklift to negotiate the test course or hitting of any markers shall constitute loss of operator control and shall constitute failure of this test.

4.6.2.12.3 Emergency steering. Test the emergency steering capability of the forklift in accordance with test method No. 10. Inability of the forklift to negotiate the test course or failure to contact all markers on the test course shall constitute inability to retain steering ability specified in 3.17.2 and shall constitute failure of this test.

4.6.2.12.4 Steering control. Test the steering control in accordance with test method No. 11. Nonconformance to 3.17.1 shall constitute failure of this test.

4.6.2.13 Highway and rail transportability. Highway and rail transportability shall be tested in accordance with test method No. 12. Nonconformance to 3.1.3.1 or 3.1.3.4 shall constitute failure of this test.

4.6.2.14 ROPS and FOPS. The ROPS shall be tested in accordance with SAE J1040. The FOPS shall be tested in accordance with SAE J1043 or SAE J231. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6.2.15 Overload. Test the forklift in accordance with test method No. 13. Nonconformance with 3.14 shall constitute failure of this test.

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4.6.2.16 Torque converter. When a torque converter is furnished, test the forklift in accordance with test method No. 14. Nonconformance to 3.11.1 or 3.25 shall constitute failure of this test.

4.6.2.17 Transmission disconnect. Test the forklift in accordance with test method No. 15. Nonconformance to 3.11.1 or 3.23.4, or exceeding the temperature limits specified for the cooling systems (see 3.10.2.6), engine oil (see 3.10), transmission oil (see 3.11), or hydraulic system (see 3.12), shall constitute failure of this test.

4.6.2.18 Service brake. Test the forklift in accordance with test method Nos. 16, 17, and 18. Nonconformance to 3.18.1 shall constitute failure of this test. For sample test (see 4.5.3.3) only test method No. 16 is required for braking.

4.6.2.19 Maintenance. Test the forklift in accordance with test method No. 19. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.20 Air transportability. Perform examination and tests detailed in MIL-STD-1791 that are applicable as determined by the Air Force, to verify air transport requirements as determined by 3.1.3.2. Nonconformance to the requirements of 3.1.3.2 shall constitute failure of this test. When contractor testing is specified (see 6.2), the government will identify a test site where the contractor shall furnish a truck for testing by government personnel.

4.6.2.20.1 External helicopter transportability. Test the forklift to the requirements of 3.1.3.3. Nonconformance to 3.1.3.3 shall constitute failure of this test. When contractor testing is specified (see 6.2), the Government will identify a test site where the contractor shall furnish a truck for testing by Government personnel.

4.6.2.21 Hydraulic system.

4.6.2.21.1 Pump inlet pressure. Test each pump in accordance with test method No. 20. Pressure differential between atmospheric pressure and the vacuum in the line in excess of that specified in 3.12.1.4 shall constitute failure of this test.

4.6.2.21.2 Hydraulic filter and filter assembly. Test the hydraulic filter in accordance with test method No. 21. Nonconformance to 3.12.3, 3.12.3.1, or 3.12.3.2, as applicable, shall constitute failure of this test.

4.6.2.21.3 Hydraulic system cleanliness. Test each truck in accordance with test method No. 22. Continue cleanup procedures until the average of not less than three consecutive counts conform to 3.12.8.

4.6.2.21.4 Hydraulic cylinders. Test each cylinder in accordance with test method No. 23. Nonconformance to 3.12.7 shall constitute failure of this test.

4.6.2.21.5 Pump contaminant tolerance. Provide certification with supporting test data for the pump 1000 hour Omega rating and the filter Beta Ten rating. Nonconformance to 3.12.2 shall constitute failure of this test.

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4.6.2.21.6 Hydraulic fittings. Test the truck in accordance with test method No. 33. The contractor shall furnish a complete set of hose assemblies with reusable fittings for conduct of this test. Nonconformance to 3.12.1.1 shall constitute failure of this test.

4.6.2.21.7 External leakage. An external leakage test shall be conducted at the conclusion of the schedule of tests indicated in table III. Observe and record accumulated leakage at practical intervals over a twelve hour period following the test(s). Leakage greater than 3.12.9 shall be cause for rejection.

4.6.2.22 Container operations. Test the forklift in accordance with test method No. 24. Nonconformance to 3.6 or 3.9.17 shall constitute failure of this test.

4.6.2.23 Sideshift, oscillation, and fork spread. Test the forklift in accordance with test method No. 25. Nonconformance to 3.13.1, 3.9.16 or 3.14 shall constitute failure of this test.

4.6.2.24 Electromagnetic interference. The forklift shall be tested in accordance with MIL-STD-461. Nonconformance to 3.28 shall constitute failure of this test.

4.6.2.25 Slinging provisions. Test the slinging provisions by applying the load for the time period specified in MIL-STD-209. Nonconformance to 3.31 shall constitute failure of this test.

4.6.2.26 Tiedown provisions. Test the tiedown provisions by applying the load for the time period specified in MIL-STD-209. Nonconformance to 3.31 shall constitute failure of this test.

4.6.2.27 Noise level test. Noise levels of first article trucks shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, Figure 7. As a minimum: noise levels shall be measured when equipment is operating under 2/3 and full load. The noise level at the typical operating position shall be provided as dB(A) level. Noise levels of production units shall be made at the operator's position and reported as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.6.2.28 Rain. Test the forklift in accordance with test method No. 26. Nonconformance to 3.4.4 or 3.10.2.8 shall constitute failure of this test.

4.6.2.29 Wheel guards. Apply a vertical downward force of 300 pounds to the geometric center of the horizontal surface, or the uppermost point (not including supports or braces), whichever is applicable. The force shall be applied for not less than 2 minutes. Examine the guard for permanent deformation or fracture. Flexible wheel guards may be deformed during the test loading but shall return to their original shape when the load is removed. Nonconformance to 3.16 shall constitute failure of this test.

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4.6.2.30 Low temperature (-25 °F). Test the forklift in accordance with test method No. 27. One or more of the following shall constitute failure of this test:

a. Damage to any part of the forklift due to storage at -25 °F (see 3.4.3).
b. Engine does not start within 5 minutes after beginning the first cranking (see 3.4.2).
c. Engine does not run smoothly within 15 minutes after starting without unnatural or continued control manipulation and without use of primers or priming aids (see 3.4.2).
d. Instruments and gauges do not operate (see 3.4.2).
e. Battery charging system does not show positive charge (see 3.4.2).
f. With the operator wearing arctic mittens and boots:

    1. Power train cannot be engaged to the wheels (see 3.4.2).
    2. Any clutch, brake, shaft, or cylinder cannot be operated through two complete movement cycles (see 3.4.2).

g. Operator encounters difficulty in mounting to and dismounting from the operator's position on the forklift (see 3.4.2).
h. Measured temperatures of engine oil (see 3.10), of engine coolant (see 3.10.2.6), or hydraulic oil (see 3.12) not within specified limits.

4.6.2.31 Fording. The forklift shall be tested in accordance with test method No. 28. Any of the following shall constitute failure of this test:

a. Evidence of water damage or malfunction of the torque converter, engine oil sump, transmission, transfer case, fuel tank, hydraulic reservoir, either axle, any wheel, instruments, or gauges (see 3.9.5).
b. Malfunction of any electrical component (see 3.9.5).
c. Malfunctioning of any brake (see 3.9.5).
d. Malfunctioning of any hydraulic component (see 3.9.5).
e. Malfunctioning of the steering system (see 3.9.5).
f. Evidence of corrosion which affects the correct functioning of the forklift (see 3.9.5).
g. Water entering the engine (see 3.10.2.9).
h. Inability to start the engine in the water (see 3.10.2.3).
i. Water entering toolbox (see 3.35).

4.6.2.32 High speed. Test the forklift in accordance with test method No. 29. Nonconformance to 3.9.19 shall constitute failure of this test.

4.6.2.33 Controls. Test the controls in accordance with test method No. 32. Nonconformance to 3.23.6 or 3.23.7 shall constitute failure of this test.

4.6.2.34 Endurance. The forklift shall be tested in accordance with test method Nos. 30 and 31. Test method No. 30 shall be conducted for not less than 200 hours. Test method No. 31 shall be conducted for not less than 250 cycles in not more than

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300 hours. Nonconformance with 3.9.18 or 3.10.2.4 shall constitute failure of this test. No endurance failures (see 4.6.2.34.1) are allowed. Relevant failures will be determined by an authorized Government representative.

4.6.2.34.1 Endurance failure. An endurance failure is any failure which results in permanent structural deformation of the truck or its components; or any failure which is corrected by redesign or by replacement of a failed component with a component of different configuration in order to meet specified performance requirements; or any failure resulting in replacement of an engine, transmission, transfer case, differential, hydraulic pump or motor, hydrostatic drive pump or motor, or the drive axle. In addition, exceeding specified temperature limits for the cooling system (3.10.2.6), engine oil (3.10), transmission oil (3.11.1), universal joints (3.11), or hydraulic system (3.12) shall constitute failure of this test.

4.6.2.35 Reliability. The forklift shall be tested in accordance with test methods numbers 30 and 31 as described in 4.6.2.34. More than 5 hardware failures or 25 unscheduled maintenance actions per vehicle shall constitute failure of this test. A hardware failure is defined as any failure which precludes the forklift from performing its intended use as described in 6.1 and which cannot be repaired within 30 minutes of occurrence by one person using on board tools and equipment. An unscheduled maintenance action is defined as any malfunction which causes degradation of specified operations or any other incident which requires action to restore the forklift to specified requirements. Relevant failures will be determined by an authorized Government representative. Nonconformance to 3.8, 3.9.6, 3.10, 3.10.2.4, 3.10.2.6, 3.11, or 3.12 shall constitute failure of this test.

4.6.2.36 Maintainability. Maintenance times shall be recorded throughout testing in accordance with 4.6.2.35. A maintenance ratio of more than 0.15 as specified in 3.7 shall constitute failure of this test.

4.6.2.37 Run-in. Each forklift shall be driven through a run-in for not less than 1 hour. The run-in shall include operating in all gears, applying the service brakes and parking brakes, using the declutch, operating all electrical equipment, and performing all hydraulic functions with rated load as well as no load. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any component, or evidence of leakage of air, fuel, lubricant or oils, shall be cause for rejection of the forklift by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after the deficiency is corrected.

4.6.2.38 Rear axle oscillation. Measure the total rear axle oscillation in accordance with test method No. 34. Nonconformance to 3.11.2 shall constitute failure of this test.

4.6.2.39 Post-test disassembly and examination. Upon completion of all endurance testing (4.6.2.34) complete the post-test list in column II of table III with the truck that underwent the endurance testing. After conducting the post-

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tests measure and record the compression in each engine cylinder and dismantle the truck sufficiently to permit detailed disassembly and visual examination of the following components:

a. Engine cylinder heads, all valves, two pistons and piston pins taken from the two cylinders having the lowest compression readings at the end of the test, one main bearing and one connecting rod bearing.
b. Fuel injection system.
c. Torque converter and transmission.
d. Universal joints and drive shafts.
e. Differential.
f. Front axle bearings and seals.
g. Complete brake system components.
h. Steering system cylinders, valves, and pump.
i. Hydraulic system pump, valves, one cylinder of each size, control levers, and any hose and fitting where leakage occurred during the test.
j. Steer axle bearings and seals.
k. Mast assembly bearings and chains.

Additional disassembly and examination shall be made at the option of the contracting officer. The truck shall be examined in accordance with table II, column 1. Record complete description of the condition of all parts and assemblies examined, and take pictures of any deficiency. Each evidence of corrosion which affects the functions of the truck or evidence of permanent deformation, breakage, or excessive wear of any component shall constitute a defect. Manufacturing tolerances listed in the contractor's quality control program shall be made available upon request and shall be used as the basis for determining excessive wear.

4.6.2.40 Nuclear-biological-chemical (NBC) protection. The forklift shall be tested in accordance with test method No. 35. Nonconformance to 3.38 shall constitute failure of this test.

4.6.2.41 DCA capability. The contractor will perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the forklift to insure wires and connections are free of error. After installation on the forklift, test the DCA harness (reference CR-82-588-003 Rev 1) on each vehicle with a DCA tester prior to interfacing with STE/ICE-R. Nonconformance to 3.42 shall constitute failure of the DCA.

4.6.2.41.1 DCA - STE/ICE-R interfaces. Upon completion of inspection and tests specified in 4.6.2.41, test the DCA for all measurements specified in 3.42 using STE/ICE-R. After completion of all first article testing, this test shall be repeated and the results compared to the first test (see 4.3.2). Nonconformance to 3.42 or failure of any componentry or wiring in the DCA harness which results in an inability to obtain the measurements previously recorded with the STE/ICE-R, shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select forklifts at any time during the contract production period and subject these forklifts to the examination specified in 4.6.1 and column 1 of table II, and to the tests marked "X" in column 1 of table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article forklifts. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison forklift to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.7, shall be cause for rejection of the inspection comparison forklift(s), and shall be cause for refusal by the Government to continue acceptance of production forklifts until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on forklifts previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all forklifts accepted prior to the completion of inspection comparison are similarly deficient, unless evidence to the contrary is furnished by the contractor, and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with the quality assurance provisions of the referenced documents in section 5 of this specification, and to determine compliance with the specified commercial preservation, packing and marking requirements.

5. PACKAGING

5.1 Preservation, packing and marking. Each forklift truck shall be preserved, packed and marked in accordance with the requirements of MIL-STD-162 for type I mobile shipment. For all degrees of preservation, liquid cooling systems shall be preserved by filling the system with equal parts of water and antifreeze conforming to MIL-A-46153. The degree of preservation shall be level A, B, C or commercial; and the degree of packing shall be level A, B, or commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The forklift is intended for use as a materials handling forklift with a capacity of loading and unloading the 8 foot wide x 8 foot high family of ANSI/ISO cargo containers, transporting, and loading boxes, containers,

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and palletized loads up to 4000 pounds. The mobility of the forklift, with and without payload, is intended to cover working areas where high flotation and traction characteristics are required.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When first article preproduction and initial production inspections are not required (see 3.2 and 6.3).
c. Time frame required for submission of the first article preproduction and initial production model(s) and number of forklifts required (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify time frame and which examination and test will be conducted by the Government, and which examination and tests shall be conducted by the contractor (see 3.2).
e. When a personnel cab is required (see 3.27).
f. When vehicles procured for Air Force shall be marked other than in accordance with MIL-STD-1223 (see 3.34).
g. When a fire extinguisher is required (see 3.5.4).
h. Degree of preservation and packing required (see 5.1).
i. When nuclear-biological-chemical (NBC) protection is not required (see 3.38).
j. When M16 mounting is not required (see 3.39).
k. When weight classification sign kit is not required (see 3.40).
l. When Government-furnished equipment (GFE) is not required (see 3.41).
m. When Government loaned equipment is not required (see 3.41.1).
n. When DCA is not required (see 3.42).
o. Color(s) required (see 3.33).
p. When toolbox must accommodate MOPP gear ensemble (see 3.35).
q. Location of air transportability and external helicopter transport test sites (see 4.6.2.20 and 4.6.2.20.1).

6.3 First article. When a first article inspections is required, the items shall be preproduction models, initial production models, or both (see 6.2). The number of vehicles for the first article inspections shall be specified by the contracting officer (see 6.2). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.3.1 Incident report. When contractor testing is required (see 6.2) the contractor shall furnish a written report within 24 hours of any incident of equipment malfunction or failure, during the conduct of any test. As a minimum, the report shall include a description of the components and parts affected, test and operating conditions, date of incident, hourmeter reading, how detected, and description of incident.

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6.4 Reserved.

6.5 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.41.

6.5.1 Government-loaned equipment. The contracting officer should arrange to loan the equipment specified in 3.41.1.

6.6 Definitions.

6.6.1 Load center. The horizontal longitudinal distance from the intersection of the horizontal load carrying surfaces and vertical load engaging faces of the forks to the center of gravity of the load.

6.6.2 Rated capacity load. Rated capacity load shall be established with a load equivalent to an unrestricted cube with overall dimensions twice the load center dimensions, and whose center of gravity is located at the geometrical center of the cube.

6.6.3 Carry position. Carry position is defined as the distance from the heel of the forks to the ground with the forks tilted fully to the rear with load centered and shall be 12 inches.

6.7 Spare parts. The contracting officer shall include provisioning requirements for spare parts.

6.8 Subject term (key word listings).

Forklift
Forklift four wheel drive
Forklift truck DED (Diesel Engine Driven)
Forklift rough terrain
4000 Pound rated load
Pneumatic Tire
Off-road operation

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - ME
Navy - SA
Air Force - 99

Review activities:

Army - AT
Navy - OS
Air Force - 84
DLA - CS

User activity:

Navy - MC

Preparing activity:

Army - ME

Project 3930-0635

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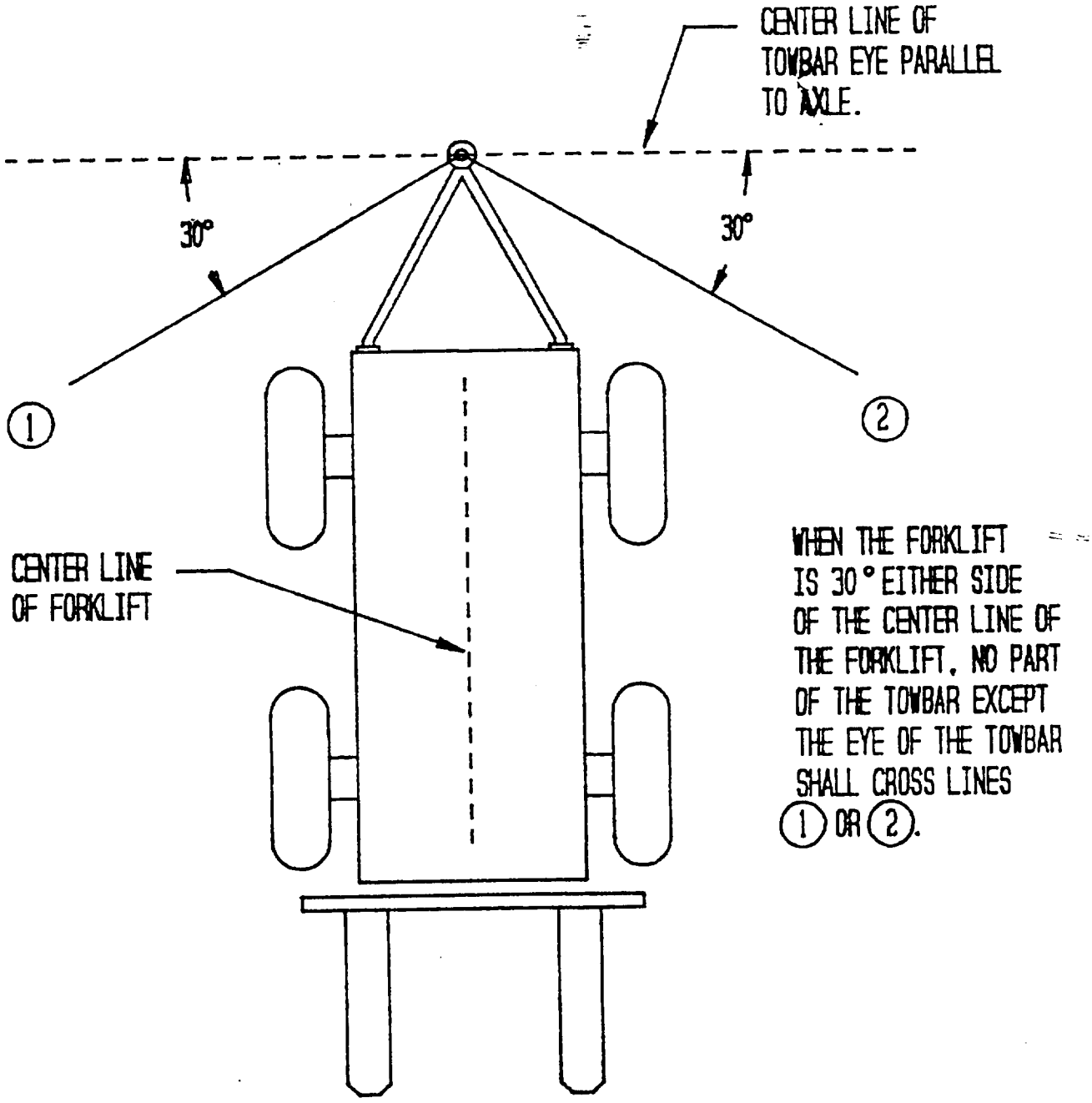
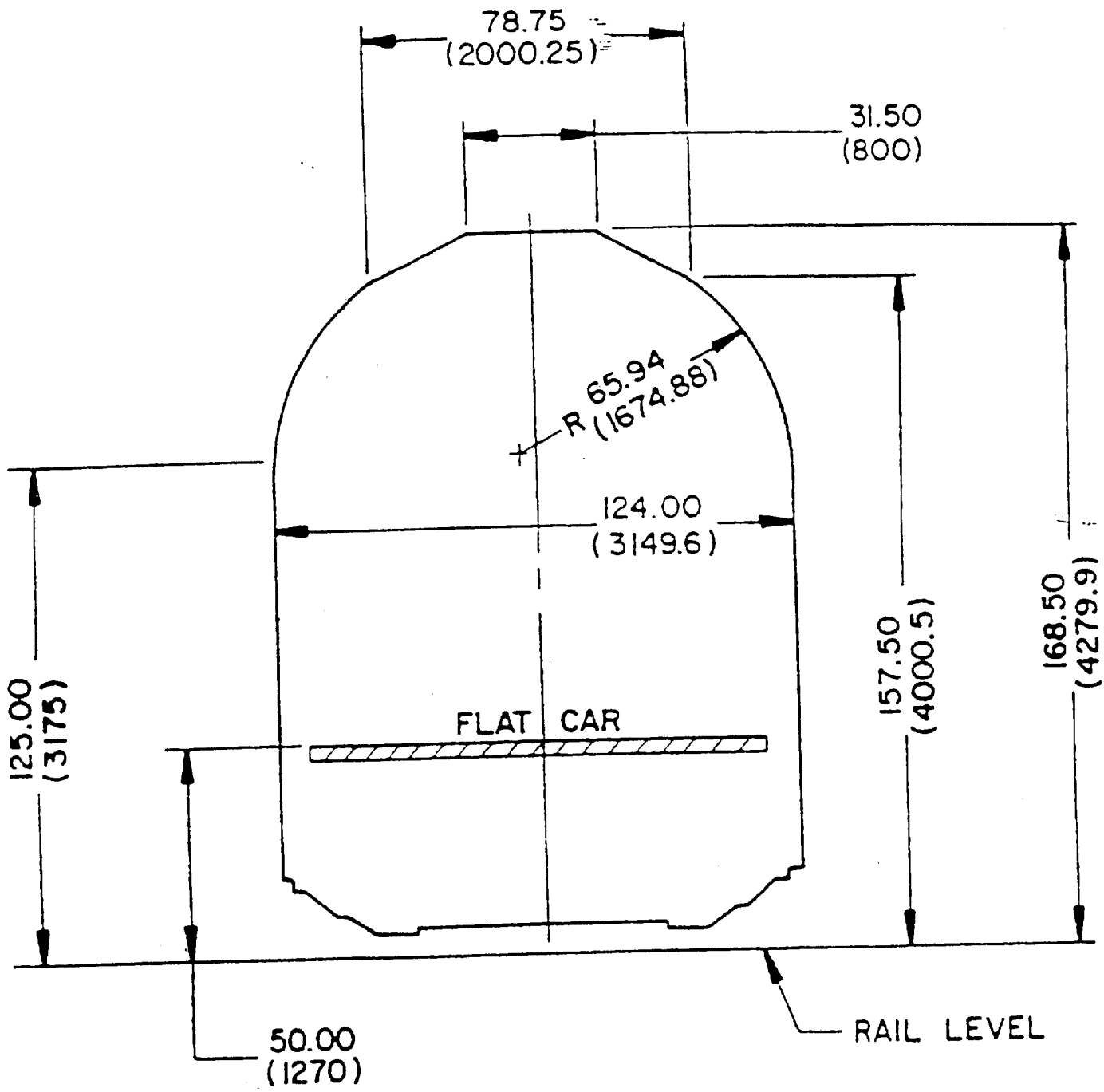


FIGURE 1: TOWBAR TURNING CLEARANCE AREA

X-29818

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NOTE: DIMENSIONS SHOWN: INCHES (MILLIMETERS)

FIGURE 2. Gabarit International De Chargement (GIC) Clearance Diagram.

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APPENDIX

FOR

TRUCKS, LIFT, FORK, DIESEL-ENGINE-DRIVEN:

PNEUMATIC-TIRED, ROUGH TERRAIN

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether forklifts procured under this specification conform to the requirements set forth.

20. APPLICABLE DOCUMENTS

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B93.28 - Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Department of Defense Materiel on Open Top Cars

(Application for copies should be addressed to the Association of American Railroads, 50 East Van Buren Street, Chicago, IL 60605.)

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APPENDIX

TEST METHOD NO. 1

WEIGHT AND DIMENSIONS

1. TEST COURSE:

   a. Level, flat surface.
   b. Level, flat surface with sharp drop-off over an edge.

2. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Liquid level or clinometer.

3. TEST PROCEDURE:

   a. Lift the rated load to the full extent of mast travel. Adjust the fork tilt to level the load. Measure the true vertical distance from the flat surface to the top surface of the forks. (This is the maximum fork height.)
   b. With mast vertical, raise the unloaded forks until the inner mast begins to exceed the specified collapsed mast height. Measure and record the true vertical distance from the flat surface to the top surface of the forks at the specified load center. (This is the free lift height.)
   c. Measure the true vertical distance from the flat surface to the uppermost projection of the upright assembly with collapsed mast in true vertical position without load as determined by a clinometer. (This is the collapsed mast height.)
   d. Position the forklift on the edge of the drop-off. Lower the forks (without load) over the edge to the full extent of mast travel. Adjust the fork tilt to level the top surface of the forks. Measure the true vertical distance between the flat surface on which the forklift is positioned and the top surface of the forks. (This is the low engagement height.)
   e. With collapsed mast in vertical position, measure the height of the unloaded truck in accordance with SAE J1234.
   f. With carriage centered, measure the width of the unloaded truck in accordance with SAE J1234.
   g. With the forks removed, measure the length of the unloaded truck in accordance with SAE J1234.
   h. Weigh the forklift fully fueled and serviced, without load. Obtain front and rear axle weights with and without rated load in carry position (see 6.6.3), for use in calculating the Flotation Index. The forklift shall be in the straight ahead position.

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APPENDIX

TEST METHOD NO. 2

LIFTING SPEED - LOWERING SPEED

1. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Stopwatch.

2. TEST PROCEDURE:

   a. Record time in seconds required to raise rated load from ground to maximum lift height (with mast vertical) at maximum engine speed.
   b. Record time to lower rated load at maximum speed (full open valve) to 3 to 4 foot height at engine idle. Load to be abruptly stopped at the 3 to 4 foot height. Record height at stop. Engine to be at low idle speed.
   c. Repeat steps "a" and "b" a minimum of 3 times.
   d. Compute lifting speed and lowering speed and use the fastest rate obtained from each as the established speeds.
   e. Remove load and raise empty forks to maximum fork height. Record time in seconds required to lower forks to lowered position at engine idle. Repeat 3 times.
   f. Compute the lowering speed and use the fastest rate as the established speed.

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APPENDIX

TEST METHOD NO. 3

TRAVEL SPEED

1. TEST COURSE:

   a. The test course shall be a dry, level surface of sufficient length to attain maximum speed (not less than 20 mph) prior to entering measured distance, plus measured distance, plus efficient stopping distance.

2. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Stopwatch or electronic timer.
   c. Pressure measurement device.

3. TEST PROCEDURE:

   a. Install exhaust system pressure measurement device 2 inches beyond the header outlet. Record exhaust pressure for each run.
   b. Drive the forklift to attain maximum speed prior to entering the measured test course. (Do not exceed an operating speed at which the forklift cannot be controlled.) While accelerating, perform gear or range changes under full throttle. Forks shall be in the carry position, (full carriage tilt backward).
   c. Drive through the measured distance at maximum speed, recording the length of the distance and the time required to traverse it. The measured distance shall be not less than 88 feet.
   d. Turn the forklift around and repeat the test approaching the measured distance from the opposite end. Repeat for a total of six runs, three in each direction through the measured distance.
   e. Conduct procedure 3a through 3d for the following:

      (1) Forward with rated load.
      (2) Rearward with rated load.
      (3) Forward without load.
      (4) Rearward without load.

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APPENDIX

TEST METHOD NO. 4

UNDERCLEARANCE, ANGLES OF APPROACH AND DEPARTURE

1. TEST COURSE:

   a. Level, flat surface.
   b. Ramp inclined to 20 degrees and a length greater than 1.2 times the truck wheelbase. At the top shall be a horizontal surface forming a sharp angle with the ramp.

2. TEST APPARATUS:

   a. Linear measurement equipment.

3. TEST PROCEDURE:

   a. Test course 1a above. Raise the rated load to carry position, (full carriage tilt backward). Measure the vertical distance from the surface to the lowest point on the vehicle, excluding wheels and tires. This is the underclearance.
   b. Test course 1b above. Drive the forklift with rated load from one horizontal surface up the ramp to the other horizontal surface in forward direction so that all wheels are on the horizontal surface. Descend the ramp in reverse. Forks may be adjusted as necessary to clear the ramp.

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APPENDIX

TEST METHOD NO. 5

UPRIGHT TILT - TILT SPEED - TILT CAVITATION -

MAST CLEARANCE

1. TEST COURSE:

a. Level, flat surface.

2. TEST APPARATUS:

a. Clinometer.
b. Stopwatch or electronic timer.
c. Rigid pole of a length not less than the width of the forklift.
d. Hydraulic pressure gauge capable of reading below atmospheric pressure.
e. Thermocouple.

3. TEST PROCEDURE:

a. Upright tilt.

(1) Place forklift without load on level, flat surface, using clinometer to insure forklift is level.
(2) Raise forks above the surface and tilt as far forward as possible and record angle on clinometer located on front surfaces of outer mast channel.
(3) Tilt forks as far rearward as possible and record angle of tilt on clinometer.

b. Tilt speed.

(1) Chain the rated load to the carriage and lift the load to maximum lift height.
(2) Tilt the load to maximum forward tilt.
(3) At governed engine speed tilt the load from maximum forward tilt, to maximum rearward tilt, and record the time required.
(4) Repeat the procedure tilting from rearward to forward at low engine idle speed. Suitable safety precautions should be observed.

c. Tilt cylinder cavitation.

(1) Install the pressure gauge to read hydraulic pressure at the head and rod ends of the tilt cylinder. Install thermocouple to read oil temperature at the hydraulic tank.
(2) After heating the oil to specified temperature, lift the rated load to maximum fork height.

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(3) With the engine at idle speed, tilt the load from full rearward to full forward and back again. Tile forward shall be done with control valve wide open 5 times and "feathered" 5 times. Observe and record any evidence of cavitation in the tilt circuit or loss of operator control.

d. Mast clearance.

(1) Position the rigid pole across the forks and anchor it as far rearward as possible on the horizontal fork surface.
(2) Operate the forks throughout the full range of lift heights and tilt angles observing any interference between the pole and any part of the forklift.

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APPENDIX

TEST METHOD NO. 6

DRIFT

1. TEST COURSE:

   a. Level, flat surface.

2. TEST APPARATUS:

   a. Temperature thermocouple and potentiometer or equivalent temperature measuring device.
   b. Clinometer.

3. TEST PROCEDURE:

   a. Install the thermocouple in the volumetric center of the hydraulic-system oil reservoir. Heat the hydraulic oil by exercising hydraulic functions until the oil temperature exceeds 120 °F. Maintain oil temperature above 120 °F for the duration of the test.
   b. Raise rated load to maximum lift height. Tilt mast to true vertical position measured with the clinometer. Mark reference point.
   c. Hold the load for 2 minutes and record the downward drift and rotational drift in that period.

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APPENDIX

TEST METHOD NO. 7

LONGITUDINAL GRADEABILITY

1. TEST COURSE:

   a. The length of the test course shall be not less than three times the wheel base of the forklift, graded to 45 percent within ±1 percent grade. Grade resistance shall be sufficient to prevent observable wheel slip. Test course surface shall be unpaved.

2. TEST APPARATUS:

   a. Speed indicator.
   b. Exhaust system pressure measurement device.

3. TEST PROCEDURE:

   a. Install exhaust system pressure measurement device not more than 2 inches beyond the header outlet. Record exhaust pressure for each run.
   b. Drive the forklift with rated load forward up the slope, entering the course at a speed of not less than 2 mph, nor more than 4 mph. Continue through the course, maintaining a speed of not less than 2 mph. Make a total of three such passes up the slope.

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APPENDIX

TEST METHOD NO. 8

DYNAMIC STABILITY - PARKING BRAKE - EMERGENCY BRAKE

1. TEST COURSE:

a. The test course shall have a length of sufficient distance to allow for 6 consecutive emergency stops and width of not less than the forklift curb clearance circle diameter. The course shall be graded to 30 percent slope, within ±1 percent grade. Test course surface shall be unpaved.

2. TEST APPARATUS:

a. Force measurement equipment.

3. TEST PROCEDURE:

a. Full circle performance.

(1) Drive the forklift with rated load centered on the forks onto the slope and perform a complete forward circle, at maximum steer angle, while on the slope.
(2) Repeat the procedure in 3a(1), steering in the opposite direction.
(3) Repeat 3a(1) and 3a(2), without load.

b. Slope descension, braking, and parking brake operation.

(1) Position the forklift with rated load at the top of the slope.
(2) Drive the forklift with rated load forward down the slope until a speed of not less than 2 mph is attained; then stop and hold the forklift with the service brakes for not less than 1 minute, with the transmission in forward gear range.
(3) Lower the load to ground level. Back away from the load until the forklift clears the load, and stop the truck with the service brake.
(4) Apply the parking brake and release the service brake. Measure and record the force necessary to apply the parking brake at the center of the handgrip. Hold for not less than 1 minute.
(5) Turn the truck around so that it is facing forward up hill. Repeat 3b(4).

c. Emergency brake operation.

(1) Position the forklift with rated load at the top of and facing down the slope. Shut off the engine and allow the forklift to roll down the slope in neutral gear range, until a speed of 2 mph is attained. Apply the emergency brake provisions (service brake) and bring the forklift to a stop. Release the brake, allow the forklift to begin to roll, and apply the brake again to stop the forklift, and repeat for a total of six consecutive dead engine stops.

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APPENDIX

TEST METHOD NO. 9

STEERING

1. TEST COURSE:

   a. The test course shall be a dry, level surface in accordance with figures A-1 and A-2.

2. TEST APPARATUS:

   a. Speed indicator.
   b. For this test only, rated capacity load (see 6.6.2) may be substituted with a 4000 lb 24-inch load center load with height at the option of the contractor.

3. TEST PROCEDURE:

   a. Obstacle course No. 1 in accordance with figure A-1. Drive the forklift forward, without load, over the test course twice, once in each direction at a speed not less than 20 mph.
   b. Repeat the procedure in "a" with rated load in carry position, (full tilt backward) at a speed not less than 20 mph.
   c. Obstacle course No. 2 in accordance with figure A-2. Drive the forklift forward, without load, over the test course, at full-load governed speed in first gear. Repeat with load in carry position.

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APPENDIX

TEST METHOD NO. 10

EMERGENCY STEERING

1. TEST COURSE:

   a. The test course shall be a dry, level surface in accordance with figure A-3.

2. TEST APPARATUS:

   a. Speed indicator.
   b. For this test only, rated capacity load (see 6.6.2) may be substituted with a 4000 lb 24-inch load center load with height at the option of the contractor.

3. TEST PROCEDURE:

   a. Drive the forklift with rated load a sufficient distance to attain a speed of not less than 20 mph. When the forklift reaches point A (figure A-3) at that speed, shut off the engine, shift the transmission lever to neutral and back into high gear. Slow the vehicle by use of the service brakes, if necessary. At point B, steer the forklift off the course, and come to a stop parallel to the course (point C). The forklift shall be under the control of the operator at all times and shall contact all of the markers on the 100 foot radius curve while negotiating the curve.

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APPENDIX

TEST METHOD NO. 11

STEERING CONTROL

1. TEST COURSE:

a. Force measurement equipment.
b. Stopwatch.
c. Linear measurement equipment.

2. TEST PROCEDURE:

a. With the engine operating at low idle, and the forklift stationary, determine the time, and the tangential force at the steering wheel rim, required to turn the wheel from stop to stop in both directions. Repeat 2 more times.
b. Repeat 2a except the engine shall operate at maximum governed speed.
c. Count the number of revolutions required to turn the wheel from stop to stop in each direction. Repeat 2 more times.
d. Apply a minimum 300 pound force downward perpendicular to the rim of the steering wheel approximately midway between two spokes and hold for a minimum of 10 seconds.
e. Repeat 2d except the force shall be applied upward.
f. With engine at low idle, apply the maximum force attainable up to 150 pounds, to the rim of the steering wheel, and hold for a minimum of 10 seconds.
g. Repeat 2f in the opposite direction.
h. Measure seat to steering wheel clearance and controls to steering wheel clearance.

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APPENDIX

TEST METHOD NO. 12

HIGHWAY AND RAIL TRANSPORTABILITY

1. TEST COURSE:

a. Paved roadway.

2. TEST APPARATUS:

a. Towing vehicle with pintle hook and speed indicator.

3. TEST PROCEDURE:

a. Highway transportability (towing).

(1) Disengage the wheels from the transmission using one man provided with common tools. Record the time necessary to perform the operation.
(2) Attach the towbar and safety chains to the towing vehicle and pull the forklift with a dead engine for not less than 50 miles at 35 mph, ±5 mph except in turns. During the 50 mile towing include a minimum of 10 right hand and 10 left hand turns at 90° intersecting roads. Observe tracking ability and note any loss of control.
(3) At the conclusion of the 50-mile-towing, disconnect the forklift from the towing vehicle, reengage the driveline and operate the forklift to observe for damage.

b. Rail impact test.

Step 1. From one to five railcars shall be positioned on a level section of track. The following conditions shall be satisfied:

(1) The total weight of the car(s) shall be at least 114,000 kilograms (250,000 pounds).
(2) The coupler between any cars shall be compressed to take up any slack.
(3) All of the air and hand brakes on the car(s) used shall be set.
(4) Any load in or on the car(s) shall be secured to prevent sliding and shifting; any movement greater than 5 centimeters (1.97 inches) shall be justification for retest.
(5) The end of the buffer car to be struck must have a standard draft gear (non-cushioned coupler).

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APPENDIX

TEST METHOD NO. 12 (Continued)

Step 2. Mount the test item on the test car. The test car shall be equipped with standard draft gears and conventional under frames. Mounting of the test item shall incorporate the standard loading and bracing method as shown in section 6 of the Association of American Railroads (AAR) "Rules Governing the Loading of Department of Defense Material on Open Top Cars." No exotic or unusual tiedown methods shall be used; any nonstandard loading and bracing must be approved by the Military Traffic and Management Command Transportation Engineering Agency (MTMCTEA), ATTN: MTT-TR, P. O. Box 6276, Newport News, VA 23606-0276 prior to testing. Substitute (dummy) test items must be of equal weight and weight distribution as the actual test item and similar in shape. The arrangement of the test item and its blocking and tiedown to be tested shall be identical to that proposed and approved by MTMTTEA (if nonstandard).

This requirement is mandatory for all equipment developed for use in the U.S. Equivalent European standards may apply for non U. S. commodities.

Step 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 65 meters (200 feet) from the buffer car(s) along a level section of track (a minimum distance to achieve the required locomotive speeds).

Step 4. Position the knuckles of the buffer and test cars for coupling.

Step 5. Install a timing device to measure the test car speed (+/- 0.1 km/hr) just prior to impact with the buffer car(s). Suggested methods include electronic timing (microswitches) and radar. The use of torpedoes and a stopwatch is permissable but not recommended because of the inaccuracies involved.

Step 6. Push the test car toward the buffer car(s) and, by using the locomotive's speedometer or other means, release the test car to freely impact the buffer car(s).

Step 7. Repeat step 6 until the test car has impacted the buffer car(s) three times at the same end, once each at speeds of 6.4, 9.7 and 13.0 km/hr (4, 6, and 8 mph) +0.8, -0.0 km/hr. Reverse the test car and repeat the 13.0 km/hr impact, for a total of 4 impacts.

NOTE: Adjustment of the lading or securing mechanisms, or reconditioning of the bracing or items of securement is not allowed without a complete retest.

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APPENDIX

TEST METHOD NO. 13

OVERLOAD

1. TEST COURSE:

   a. The test course shall be a flat, level surface with provisions for anchoring the forklift to prevent tipping and for blocking the front axle to prevent overloading the tires.

2. TEST APPARATUS:

   a. Test loads or other apparatus to apply load to forks.

3. TEST PROCEDURE:

   a. Anchor the forklift to prevent tipping and block the front axle to prevent overloading the tires. Raise the forks to provide not less than 8 inches ground clearance to level forks.
   b. Center the forks and apply three times the rated load on the forks at 24 inch load center distance, and hold for 5 minutes.
   c. Sideshift the forks fully to the right and apply two times the rated load on the forks at 24 inch load center distance, and hold for 5 minutes.
   d. Sideshift the forks fully to the left and repeat the procedure in "c".
   e. Remove the load and examine the forklift for damage and hydraulic leaks.

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APPENDIX

TEST METHOD NO. 14

TORQUE CONVERTER - PINTLE HOOK

1. TEST APPARATUS:

   a. Timing device.
   b. Thermocouple and potentiometer or equivalent temperature measuring device.
   c. Tachometer.
   d. Thermometer.

2. TEST PROCEDURE:

   a. Install thermocouple to measure oil temperature at the outlet from the converter.
   b. Thermostat shall not be blocked open.
   c. With the ambient temperature not less than 25 °F, operate the forklift until the torque converter oil outlet temperature is a minimum of 157 °F.
   d. Attach the forklift pintle hook to a fixed object to prevent forward motion of the forklift by a horizontal force on the pintle hook. Shift into the highest forward gear and stall the torque converter for not less than 30 seconds while the engine is operating at governed speed. The transmission disconnect control shall not be used during this test.
   e. Record initial torque converter oil temperature and maximum oil temperature after 30 seconds stall, governed speed at stall, and ambient temperature.

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APPENDIX

TEST METHOD NO. 15

TRANSMISSION DISCONNECT

1. TEST APPARATUS:

   a. Thermocouples and potentiometers or equivalent temperature measuring device.
   b. Timing device.
   c. Linear measurement equipment.
   d. Pedal force gauge.

2. TEST PROCEDURE:

   a. Install thermocouples to record temperatures listed below:

      (1) Engine outlet to top tank of the radiator.
      (2) Engine oil gallery.
      (3) Transmission sump oil to cooler.
      (4) Transmission oil from cooler.
      (5) Hydraulic tank.
      (6) Ambient air (thermometer may be used).

   b. Block open thermostat.
   c. Place truck on level surface with forks elevated approximately 1 foot from the surface.
   d. Measure the pedal force required to activate the declutch control.
   e. With direction control in forward position, and the gear selector, if applicable, in 1st gear, depress accelerator to provide governed speed of engine and at the same time actuate the declutch pedal to maintain a speed of 1/2 to 1-1/2 miles per hour. Continue for not less than 20 minutes. Record temperatures at the beginning of the period and at 5 minute intervals thereafter. Speed shall be computed from the total distance traveled over the 20 minute period.
   f. With transmission disconnect pedal fully depressed, transmission in forward low range, continually raise and lower rated load for a minimum of 5 minutes while engine is at governed speed. Record initial and final temperatures and pedal force required to depress pedal.

Note: Oil temperatures shall not exceed 30 °F above ambient at the start of the tests in 2e and 2f.

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APPENDIX

TEST METHOD NO. 16

SERVICE BRAKE

1. TEST COURSE:

a. The test course shall be a dry, level, improved surface allowing no penetration, with a grade not exceeding ±1 percent.

2. TEST APPARATUS:

a. Linear measurement equipment.
b. Detonator, brake apparatus, with powder loads, or equivalent apparatus.
c. Speed indicator.
d. Pedal force gauge.

3. TEST PROCEDURE:

a. Stopping distance.

(1) Attach the pedal force gauge to the brake pedal in a manner which enables force applied to the pedal to be measured.
(2) Connect the detonator to be operated by a switch directly connected to the brake pedal so that it is actuated during the first 1/4 inch of pedal travel.
(3) Place the rated load on the forks in a carry position (see 6.6.3).
(4) Drive the forklift with rated load over the test course. When the forklift reaches minimum 15 mph, apply the brakes. Measure and record the pedal force and the distance between the point of brake pedal application and the point of stoppage. Perform the test four times in forward gear; two times in each direction on the course. Observe whether the front wheels lock or any wheel loses contact with the test course. Measure the amount of fishtail.

b. Brake system proof.

(1) Apply a 250 pound force to the pedal, hold for 1 minute and release. Repeat this operation for a total of 10 times.

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APPENDIX

TEST METHOD NO. 17

BRAKE FADE

1. TEST COURSE:

   a. The test course shall be a dry, improved surface allowing no penetration, with grades not exceeding ±1 percent.

2. TEST APPARATUS:

   a. Speed indicator.
   b. Decelerometer.

3. TEST PROCEDURE:

   a. Drive the forklift with rated load up to a minimum of 15 mph and apply the service brakes to produce a deceleration rate of 8 fps per second, ±1 fps per second, until the forklift has slowed to a minimum of 5 mph. Immediately accelerate the forklift at full throttle in high gear until the travel speed reaches a minimum of 15 mph. Immediately repeat the test for a total of 35 cycles. On completion of the 35th cycle, accelerate the forklift at full throttle in high gear to a minimum 15 mph and immediately conduct a service brake stopping test using procedure in test method No. 16.
   b. After the brakes have completely cooled, perform 3a again. Make no adjustments or repairs between the two test runs.

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APPENDIX

TEST METHOD NO. 18

WET BRAKE

1. TEST COURSE:

   a. Body of fresh water of sufficient depth to permit submersion of the forklift to a minimum 20 inch depth.

2. TEST PROCEDURE:

   a. Drive the forklift with rated load into the water to the minimum 20 inch depth. Release the service brakes and allow the forklift to stand for not less than 15 minutes. After the 15 minutes have elapsed, drive the forklift from the water and immediately conduct a service brake stopping test, using the procedure specified in 3a of test method No. 16. Service brakes shall not be applied after release in the water until the test application.

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APPENDIX

TEST METHOD NO. 19

MAINTENANCE

1. TEST FACILITY:

   a. The test facility shall consist of an area of sufficient size to permit unobstructed surveillance of maintenance.

2. TEST APPARATUS:

   a. Common handtools.
   b. Simplified Test Equipment for Internal Combustion Engines Reprogramable (STE/ICE-R).

3. TEST PROCEDURE:

   a. Examine the forklift for accessibility of major assemblies.
   b. Examine position of drains with respect to accessibility.
   c. Using the tools, demonstrate that the major assemblies are accessible for repair and maintenance.
   d. Demonstrate the accessibility of drains and path of discharge of lubricants and coolants by activating drainage controls.
   e. Perform and time each operation on the maintenance operations list (see 3.7).
   f. Connect the STE/ICE-R to record all parameters which are to be measured in the Transducer Kit (TK) mode.

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APPENDIX

TEST METHOD NO. 20

PUMP INLET PRESSURE

1. TEST APPARATUS:

   a. Pressure gauge.
   b. Hydraulic fittings (as required).
   c. Thermocouple.

2. TEST PROCEDURE:

   a. Install a pressure (vacuum) gauge in the pump suction line immediately adjacent to the pump (or at the pump) and a thermocouple in the hydraulic tank.
   b. Heat the oil to specified temperature.
   c. Run the pump at the maximum speed it can encounter during vehicle operation and record the pump-inlet pressure, and temperature.
   d. Repeat for each pump installation on the truck.

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APPENDIX

TEST METHOD NO. 21

HYDRAULIC FILTER AND FILTER ASSEMBLY

1. TEST APPARATUS:

a. Hydraulic test stand may be used in place of on-vehicle test.
b. Pressure gauges.
c. Temperature gauges.
d. Flowmeter (when test stand is used).
e. Provisions for controlling oil temperature, if necessary.

2. TEST PROCEDURE:

a. Pressurize the housing to not less than the minimum proof pressure (100 psi for return line filters, 30 psi for suction line filters, or 1-1/2 times maximum system pressure for pressure filters). Hold the pressure for not less than 60 seconds and examine for leaks.
b. Install a new element in the filter housing. Establish maximum flow through the filter at an oil temperature of 150 ±5 °F. Measure and record the pressure drop across the filter assembly.
c. Plug the filter element chamber oil exit, or otherwise obstruct all flow through the filter element, leaving the outlet port open. Establish and maintain the oil temperature at 150 ±5 °F. Increase the upstream pressure to 10 psi more than determined in "b" or to 2 psi for suction line filters. Observe the contamination indicator. Increase upstream pressure until the indicator enters the bypass area or until bypass flow exceeds 5 percent of the maximum system flow, whichever occurs first. Observe and record this pressure. (This is the bypass opening pressure.) Increase pressure until the maximum system flow is attained through the bypass valve. Observe and record this pressure.

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APPENDIX

TEST METHOD NO. 22

HYDRAULIC SYSTEM CLEANLINESS

1. TEST APPARATUS:

   a. Liquid automatic particle counter.

2. TEST PROCEDURE:

   a. Connect the hydraulic lines at the cylinders to bypass those cylinders whose capacity is less than 1/2 of the connecting line volume.
   b. Run the engine at no load governed speed pumping the oil through each circuit for not less than 5 minutes. Perform the particle count as described in "d" below.
   c. When applicable, connect each line to its related component. Operate each function of the system for 5 minutes continuously through full stroke and at maximum rates. Perform the particle count as described in "d" below.
   d. The procedure for determining contamination shall be as follows:

      (1) Contamination levels shall be determined by particle counts. The particle counter used for evaluation of the samples shall be calibrated in accordance with ANSI Recommended Standard Method for Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust (ANSI B93.28). The required counts shall be the average of not less than three consecutive counts. Samples may be taken from the reservoir or upstream of the filter(s) used for cleanup.
      (2) Use any combination of on-the-forklift and off-the-forklift filtration for cleanup.
      (3) Connect the counter sensor to the system, using a bypass line and a branch line. The bypass line shall connect to the system, have a flow rate not less than 5 times the sensor flow rate, and may return fluid directly to the reservoir. The branch line to the sensor shall tee into the bypass line and shall be not more than 12 inches in length. Any flow restriction devices shall be placed downstream of the particle counter sensor. Start engine and operate at not less than 1000 rpm. Record particle counts.

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APPENDIX

TEST METHOD NO. 23

CYLINDER PROOF

1. TEST APPARATUS:

a. Hydraulic test bench or the truck capable of developing two times system working pressure.
b. Pressure gauge.
c. Thermocouple.

2. TEST PROCEDURE:

a. Double acting cylinder.

(1) Position and mechanically hold the piston at the approximate midpoint of the cylinder.
(2) Fill both sides of the piston with oil at a minimum of 120 °F.
(3) With the head end port capped, apply an oil pressure equal to 1.5 times the working pressure to the rod end of the piston for a minimum of 60 seconds. Release pressure.
(4) With the rod end port capped, apply an oil pressure equal to 1.5 times the working pressure to the head end of the piston for a minimum of 60 seconds. Release pressure.

b. Single acting cylinder.

(1) Extend cylinder to maximum extension by filling with oil at a minimum of 120 °F.
(2) Apply oil pressure equal to 1.5 times the working pressure at the inlet port for a minimum of 60 seconds. Release pressure.

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APPENDIX

TEST METHOD NO. 24

CONTAINER OPERATION

1. TEST FACILITY:

   a. The test facility shall consist of a flat, level area of sufficient size to permit operation of the forklift around the container, ramp, and trailer chassis.

2. TEST APPARATUS:

   a. Container MH5.1.1m, 20 feet long x 8 feet wide x 8 feet high.
   b. Ramp: Loading vehicle in accordance with MIL-R-52899.
   c. Container chassis minimum 55 inch height.

3. TEST PROCEDURE:

   a. Place the container on the chassis or platform. Place the end of the ramp into the container. Drive the forklift up the ramp and all the way into the container. Operate the forklift sideshift and raise forks to free lift height while in the container. Back the forklift out of the container and down the ramp to ground level. Check for interferences between the truck and container while entering the container, operating the container, and exiting the container. Check for visibility of the ramp curb in the area of the left rear wheel of the truck when backing out of the container.

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APPENDIX

TEST METHOD NO. 25

SIDESHIFT, OSCILLATION, AND FORK SPREAD

1. TEST APPARATUS:

a. Clinometer.
b. Linear measurement equipment.
c. Stopwatch.

2. TEST PROCEDURE:

a. Clockwise oscillation.

(1) With the forklift on a level surface, place rated load on the forks and sideshift the load to extreme left side position. Place a clinometer on the load or carriage.
(2) Oscillate the load to extreme clockwise position and record the angle of oscillation relative to horizontal obtained.
(3) Oscillate the load to clockwise position and record the angle of oscillation relative to horizontal obtained.
(4) Repeat 2a(2) and 2a(3) at low engine idle and high engine idle a minimum of 3 times each and record time for each.

b. Counterclockwise oscillation.

(1) With the forklift on a level surface, place rated load on the forks and sideshift the load to extreme right side position. Place a clinometer on the load or carriage.
(2) Oscillate the load to extreme clockwise position and record the angle of oscillation relative to horizontal obtained.
(3) Oscillate the load to extreme counterclockwise position and record the angle of oscillation relative to horizontal obtained.
(4) Repeat 2b2 and 2b3 at low engine idle and at high engine idle a minimum of 3 times and record time for each.

c. Forks and fork spacing.

(1) With no load on the forks, move the forks as close together as possible and measure the distance between the inside edges of the forks. Move the forks as far apart as possible and measure the distance over the outside edges of the forks.
(2) Measure the fork length, fork width, fork thickness at the heel and at the tip, and the fork taper.

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d. Sideshift.

(1) Spread the forks to a minimum 34 inches (outside to outside).
(2) Sideshift level forks to extreme left position. Note and measure any increase in truck width and whether there are any protrusions beyond the sideshift carriage.
(3) Sideshift forks to extreme right position. Measure sideshift travel. Note and measure any increase in truck width and whether there are any protrusions beyond the sideshift carriage.
(4) Sideshift forks to extreme left position and measure travel.
(5) Place rated load on the forks. Sideshift to the extreme right position. Oscillate the load to extreme counterclockwise position.
(6) Oscillate the load to the extreme left side position. Oscillate the load to the extreme counterclockwise position.
(7) With rated load on the forks in the horizontal position, operate the sideshift function for 100 full cycles at maximum speed. During the last 50 cycles, the sideshift control shall be released abruptly when the load is in approximately midposition, and the cycle shall be continued.

Notes: Sideshift cycles may be operated in 10 cycle intervals to allow sufficient cooling of the system.

One cycle is defined as sideshifting from extreme left to extreme right and return to extreme left.

All directions are determined relative to the operator sitting forward in the operator's seat.

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APPENDIX

TEST METHOD NO. 26

RAIN

1. TEST APPARATUS:

   a. Timing device.
   b. Measuring device (U.S. Weather Bureau type preferred).
   c. Spray fixture.

2. TEST PROCEDURE:

   a. Drain a small quantity of fluid from each fluid's reservoir through the drain plug (except cooling system) and visually examine to ensure that the water is not present in the fluids before conducting the test.
   b. Subject the forklift to a simulated rainfall of not less than 4 inches per hour. The simulated rainfall shall impinge on the forklift at an angle of 45 degrees, ±5 degrees, with a force equivalent to a 30 mile per hour wind. The simulated rainfall shall disperse uniformly over the surfaces of the forklift.
   c. Expose the top and left side of the forklift to the rain described above for a period of 5 minutes with engine off. With rain still flowing, start engine and allow engine to idle for 2 minutes.
   d. Repeat "b" with the top and front exposed.
   e. Repeat "b" wit top and right exposed.
   f. Repeat "b" with top and rear exposed.
   g. Stop the rainfall and allow the forklift to stand for not less than 30 minutes with the engine off. Drain a small quantity of fluid from each fluid's reservoir through the drain plug (except cooling system) and visually examine for presence of water.
   h. Visually check electrical components, gauges, cab (if furnished), and the air cleaner for any water entry or damage.

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APPENDIX

TEST METHOD NO. 27

LOW-TEMPERATURE TEST (-25 °F)

1. TEST APPARATUS:

a. Cold chamber.
b. Arctic clothing.
c. Thermocouples with potentiometer(s) or other temperature measuring devices.
d. Battery hydrometer.
e. Wattmeter or ammeter.
f. Voltmeter.

2. TEST PROCEDURE:

a. Service the forklift with applicable lubricants for -25 °F operation including engine oil, transmission oil, drivetrain lubricants, hydraulic oil and antifreeze.
b. Check the specific gravity of battery electrolyte to insure full charge.
c. Cold soak the forklift to attain a stabilized temperature no warmer than -25 °F. Stabilization is attained when the engine coolant, lubricating oil, and battery temperatures reach -25 °F, as indicated by the thermocouples located as follows:

(1) In the two center cells of each battery between the plates and at middepth of the electrolyte.
(2) In the volumetric center of the hydraulic system oil reservoir.
(3) In the engine lubricating oil gallery and in the oil sump.
(4) In the engine coolant in the engine cooling jacket discharge pipe.
(5) In the chamber at a horizontal angle of 45 degrees from the sides and ends of the truck, not less than 3 feet from each corner of the truck at engine crankshaft level.

d. Maintain -25 °F, ambient temperature throughout the entire test.
e. One operator wearing arctic clothing, including mittens, shall perform all priming and starting operations.
f. Substitute or supplementary batteries or other electrical cranking energy shall not be used.
g. Glowplugs or other accessories working off the cranking batteries and auxiliary fluid priming systems which are permanently mounted on the forklift and normally used in starting the engine may be used.

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h. Start the forklift. Starting should be accomplished within 5 minutes of the initial attempt. Smooth engine running without unnatural or continued control manipulation, and without use of primers and priming fluids should be obtained within 15 minutes after starting.

i. When started, run engine 15 minutes. After completion of the 15 minute operation, operate the engine as follows:

(1)	10 minutes idling.
(2)	10 minutes half throttle.
(3)	5 minutes full throttle.

Perform the 25 minute cycle three times.

j. Record temperatures prior to engine starting, immediately after engine starting, and at 10 minute intervals during warmup and cycling, including but not limited to the following:

(1)	Battery.
(2)	Hydraulic tank.
(3)	Engine oil.
(4)	Engine coolant.
(5)	Chamber air.

k. During engine cycling, and while wearing arctic mittens and boots, the operator shall perform each of the following:

(1)	Engage and disengage power train to the wheels in all gears.
(2)	Operate through two complete movement cycles all brakes, shafts and cylinders (as permitted by space limitations in chamber).
(3)	Mount to and dismount from the operator's position on the forklift.

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APPENDIX

TEST METHOD NO. 28

FORDING

1. TEST COURSE:

a. The test course shall be a body of water of sufficient depth to permit submersion of the forklift to a minimum 20-inch depth.

2. TEST PROCEDURE:

a. Drain a small quantity of fluid from each fluid's reservoir through the drain plug (except cooling system) that will be in contact with the water and visually examine to ensure that water is not present in the fluids before the test.

b. Drive the forklift with rated capacity load into the water to the minimum depth of 20 inches. Stop the truck in the water by applying the service brakes. Maneuver the forklift back and forth by shifting the transmission from forward to reverse, a minimum of 5 times. Before the direction of the forklift is changed the service brakes shall be applied to stop the forklift. During the maneuvering the steering system shall be turned from lock to lock a minimum of 5 times. Drive the forklift out of the water. During the test the load may be raised to keep it out of the water.

c. Repeat step 2a for a total of five times.

d. Drive the forklift without load on the forks, in the carry position, into the water to a minimum depth of 20 inches. Stop the forklift in the water by applying the service brakes. Operate all of the hydraulic cylinders, except steering cylinders, through one complete movement cycle. Shut the engine down and restart it. Maneuver the forklift back and forth by shifting the transmission from forward to reverse a minimum of 5 times. Before the direction of the truck is changed the service brakes shall be applied to stop the forklift. During the maneuvering the steering system shall be turned from lock to lock a minimum of 5 times. Drive the forklift out of the water.

e. Repeat step 2c for a total of five times.

f. Drain a small quantity of fluid from each fluid's reservoir through the drain plug (except cooling system) that were in contact with the water and note any visual evidence of water in the liquids drained from the reservoir.

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APPENDIX

TEST METHOD NO. 29

HIGH SPEED

1. TEST COURSE:

   a. The test course shall be a compacted surface of sufficient size to allow continuous travel at wide open throttle in highest gear range.

2. TEST APPARATUS:

   a. Speed indicator.
   b. Equivalent temperature measuring device.

3. TEST PROCEDURE:

   a. Block thermostat in the open position.
   b. Install thermocouples in the hydraulic system reservoir near the pump inlet, in the transmission sump, in the transfer case sump, in the engine coolant outlet, in the torque converter outlet line adjacent to the torque converter and engine oil gallery.
   c. Record all temperatures including ambient before starting test.
   d. Drive the forklift over the test course for 2 hours at wide open throttle in highest forward gear range. Stop the forklift and record all temperatures immediately, including ambient and universal joint temperatures. Check for universal joint temperature using temperature marker.
   e. In not more than 5 minutes after stopping, drive the forklift for an additional 2 hours as in "d", and record all temperatures immediately upon stopping, including ambient and universal joint temperatures.

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APPENDIX

TEST METHOD NO. 30

PERFORMANCE (DURABILITY COURSE)

1. TEST COURSE:

   a. A test course shall be constructed in accordance with figure A-4.

2. TEST APPARATUS:

   a. Themocouples and potentiometer(s) or equivalent temperature measuring device.

3. TEST PROCEDURE:

   a. Block thermostat in the open position.
   b. Install thermocouples in the hydraulic system reservoir near the pump inlet, in the transmission sump, in the transfer case sump, in the engine coolant outlet, in the torque converter outlet line adjacent to the torque converter, and the engine oil gallery. One day per week, measure the temperatures, including ambient, at successive hourly intervals for a period of 6 hours.
   c. Drive the forklift, with lights on, with rated load in carry position (see 6.6.3). The speed throughout the entire course shall be not less than 3.5 mph or greater than 4.0 mph. For this test, the engine governor setting may be changed or accelerator stop may be installed to maintain speed throughout the test. The operator shall not vary the accelerator position while negotiating the test course between points C and D.
   d. One cycle is completed each time the forklift travels from loop A to loop B and returns to loop A. Turn around in loop A shall consist of right hand turns and turn around in loop B shall consist of left hand turns.

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APPENDIX

TEST METHOD NO. 31

PERFORMANCE (CONTAINER/TRAILER LOADING)

1. TEST COURSE:

a. A test course shall be constructed in accordance with figure A-5.

2. TEST APPARATUS:

a. Thermocouple and potentiometer or equivalent temperature measuring device.
b. Palletized loads, 16 each, mounted on 40 by 48 inch pallets, load enclosed in a box between 41 and 43 inches overall height including pallet. The gross weight of the pallets shall be: Six 4000-pound, five 2500-pound, and five 1000-pound, ±100 pounds each. The pallets shall be suitable for stacking up to three high, and shall provide adequate clearance underneath to be engaged by the forks, but not in excess of 6 inches.
c. Container MH5.1.1m, 20 feet long x 8 feet high, mounted on a trailer chassis with auxiliary support for the fifth wheel.
d. Container MH5.1.1m, 20 feet long x 8 feet high, on the ground.
e. Trailer, highway type, flatbed large enough to carry all 16 pallets, not stacked.
f. Ramp: Loading vehicle in accordance with MIL-R-52899.
g. Fence or barricade 60 feet long.

3. TEST PROCEDURE:

a. Block thermostat in the open position.
b. Install thermocouples in the hydraulic system reservoir near the pump inlet, in the transmission sump, in the transfer case sump, in the engine coolant outlet, in the torque converter outlet line adjacent to the torque converter, and the engine oil gallery. One day per week measure the temperatures, including ambient, at successive hourly intervals for a period of 6 hours.
c. Handle all pallets from the 40 inch end. Start the test with all pallets in the trailer mounted container (four rows of two 2-high stacks).
d. Move all pallets to the pallet stack area, stacking 3 high (five 3-high stacks and one not stacked). When stacking, the forklift (except forks) shall remain between the barricade and the marked boundary. The forklift tires shall not cross the marked boundary during any portion of the test.
e. Move all pallets to the flatbed trailer, loading from both sides of the trailer and not stacking (two parallel rows of 8 pallets). The first pallet moved to the trailer shall not be the last one that was placed in the pallet stack.

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APPENDIX

TEST METHOD NO. 31 (Continued)

f. Move all pallets to the container on the ground, stacking as in "c". The first pallet moved to the container shall not be the last one that was placed on the trailer.
g. Move all pallets to the trailer mounted container, stacking as in "c". The first pallet moved to the trailer mounted container shall not be thee last one that was placed in the container on the ground.
h. Steps "d" through "g" constitutes one cycle. A cycle is completed each time the sixteenth pallet is loaded into the trailer mounted container.

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APPENDIX

TEST METHOD NO. 32

CONTROLS

1. TEST APPARATUS:

   a. Thermocouples and potentiometer or equivalent temperature measuring device.
   b. Force measurement equipment.
   c. Linear measurement equipment.

2. TEST PROCEDURE:

   a. Install the thermocouple in the hydraulic reservoir.
   b. With an oil temperature of 140 to 180 °F and the engine operating at no load governed speed, measure the force and distance required to shift each hydraulic control valve at the handgrip to each position.
   c. Apply 100 pounds of force to each valve lever in each shifting direction and maintain the force for not less than 15 seconds.
   d. Apply 100 pounds of force to each transmission control lever in each direction and maintain the force for not less than 15 seconds.

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APPENDIX

TEST METHOD NO. 33

HYDRAULIC FITTINGS

1. TEST APPARATUS:

   a. Common handtools.

2. TEST PROCEDURE:

   a. Remove one hydraulic hose assembly that has permanent-type fittings.
   b. Replace the hose assembly with an equivalent field attachable fitting only to assure interchangeability.
   c. Repeat "a" and "b" above until each hose assembly with permanent-type fittings is replaced with a hose assembly with reusable fittings.
   d. After every hose assembly with permanent-type fittings has been replaced with a hose assembly with reusable fittings, reinstall the original hose assemblies on the truck.

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APPENDIX

TEST METHOD NO. 34

REAR AXLE OSCILLATION

1. TEST APPARATUS:

   a. Clinometer.

2. TEST PROCEDURE:

   a. Raise the forklift, keeping the forklift horizontal, to sufficient height to allow the rear axle to oscillate from stop to stop.
   b. Attach the clinometer to the rear axle.
   c. Oscillate the rear axle fully clockwise. Measure and record the angle from the horizontal.
   d. Oscillate the rear axle fully counterclockwise. Measure and record the angle from the horizontal.

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APPENDIX

TEST METHOD NO. 35

NUCLEAR-BIOLOGICAL-CHEMICAL

1. TEST COURSE:

a. The test course shall be constructed in accordance with figure A-5.

2. TEST APPARATUS:

a. Palletized loads, 16 each, mounted on 40 by 48 inch pallets, load enclosed in a box between 41 and 43 inches overall height including pallet. The gross weight of the pallets shall be: Six 4000-pound, five 2500-pound, and five 1000-pound, ±100 pounds each. The pallets shall be suitable for stacking up to three high, and shall provide adequate clearance underneath to be engaged by the forks, but not in excess of 6 inches.
b. Container, MH5.1.1m, 20 feet long x 8 feet high, mounted on a trailer chassis with auxiliary support for the fifth wheel.
c. Container, MH5.1.1m, 20 feet long x 8 feet high, on the ground.
d. Trailer, highway type, flatbed large enough to carry all 16 pallets, not stacked.
e. Ramp: Loading vehicle in accordance with MIL-R-52899.
f. Fence or barricade, approximately 4 feet high by 60 feet long.
g. The complete protective ensemble.

3. TEST PROCEDURE:

a. While wearing the complete protective ensemble, each and every control (i.e., levers, buttons, switches, pedals, knobs, and handles) shall be actuated by the operator. Record any incident of interference caused by the ensemble that is detrimental to the operation of the forklift.
b. Complete steps 3c to 3g as stated in test method no. 31.
c. The operator shall not be given any assistance in engaging any loads during this test. The forklift shall maneuver within the area of the course during all operation.
d. Record any detrimental effects the ensemble has on the operational capabilities of the forklift and record the time for the cycle.

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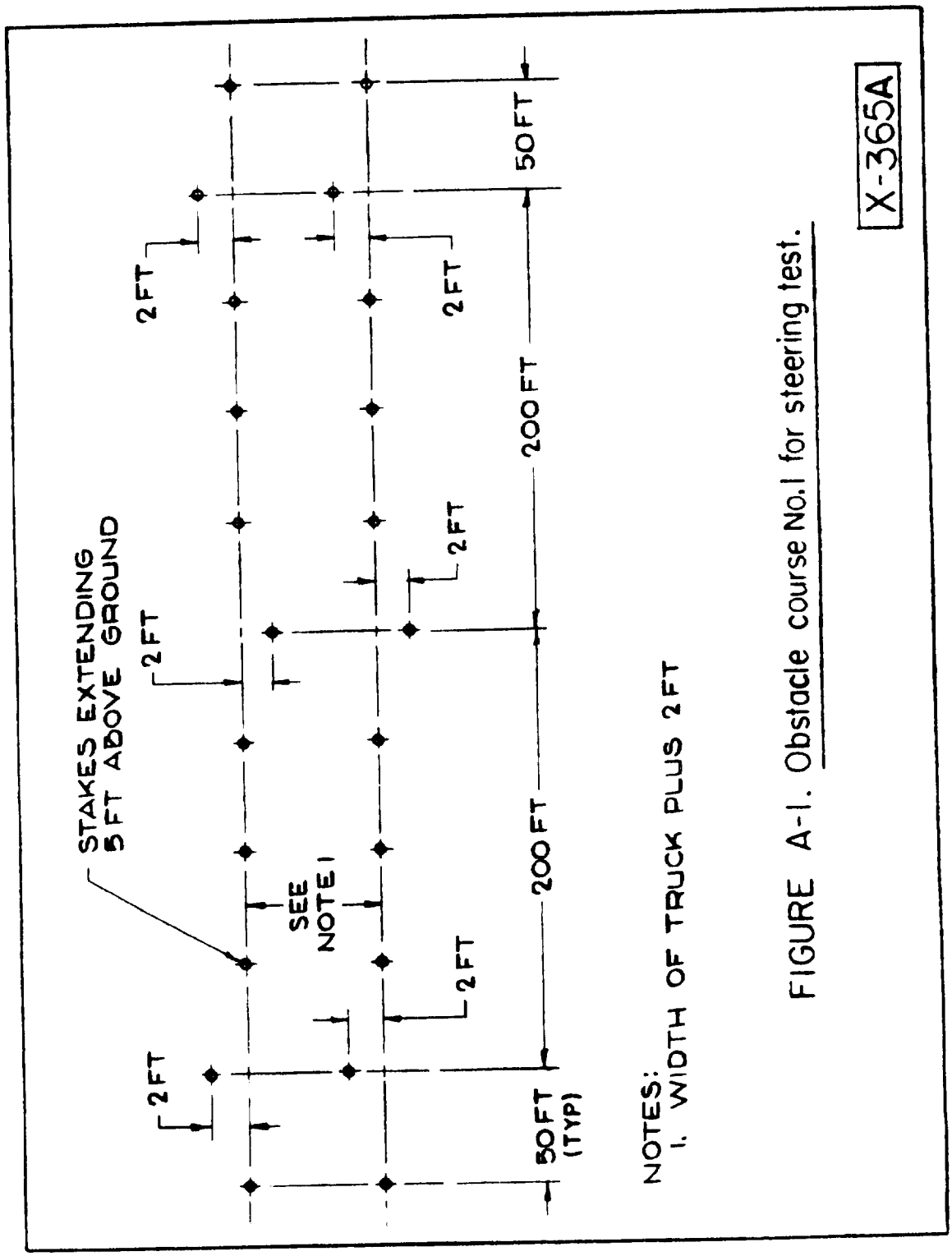
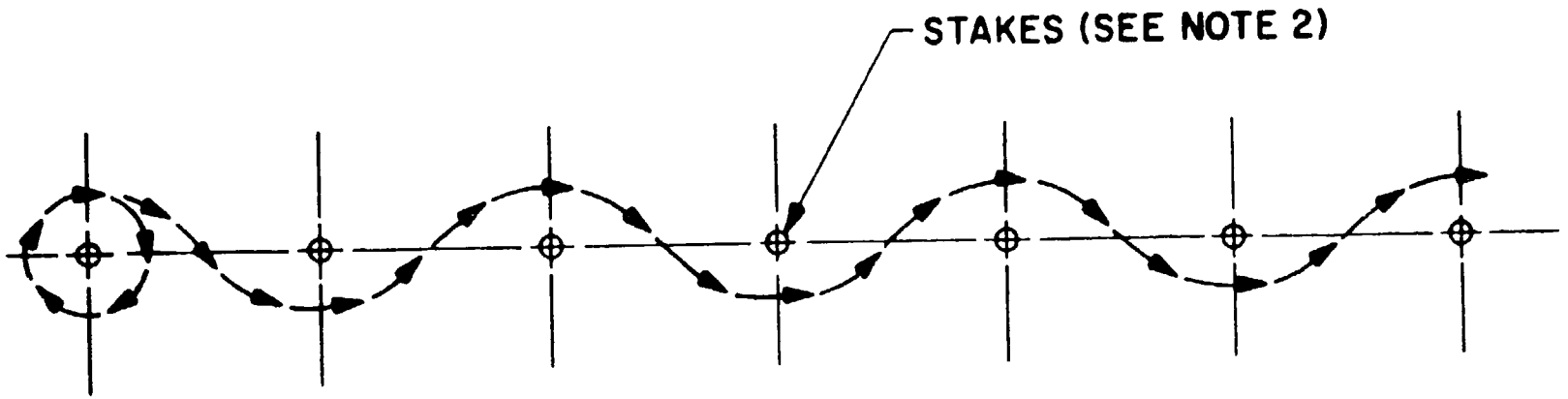


FIGURE A-1. Obstacle course No.1 for steering test.

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(SEE NOTE 1)

NOTES:

1. THE SPACE BETWEEN THE STAKES SHALL BE EQUAL TO TWO TIMES THE CURB CLEARANCE CIRCLE DIAMETER.
2. STAKES SHALL BE HIGH ENOUGH TO BE EASILY SEEN BY THE OPERATOR WHILE GOING THROUGH THE TEST COURSE.

FIGURE A-2. Obstacle No. 2 for course steering test.

X-2983E

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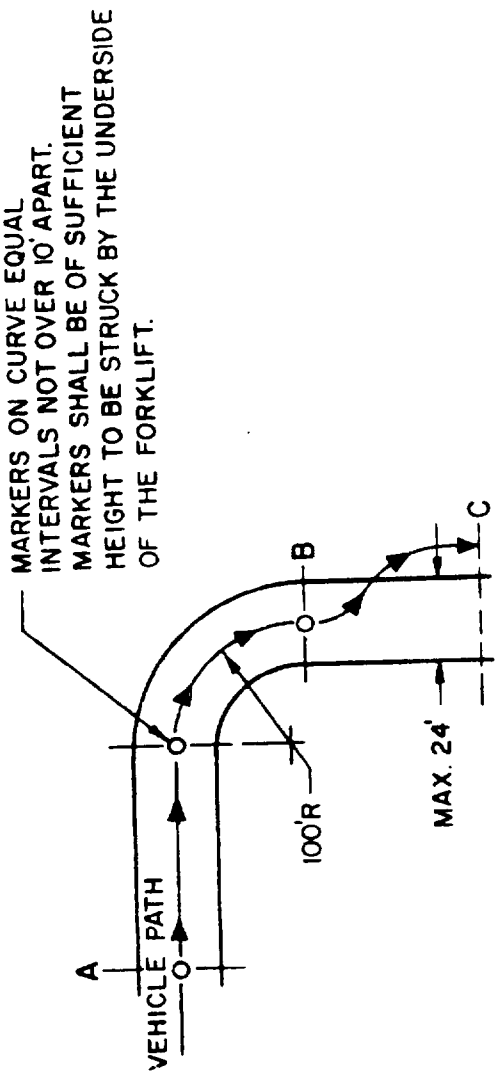
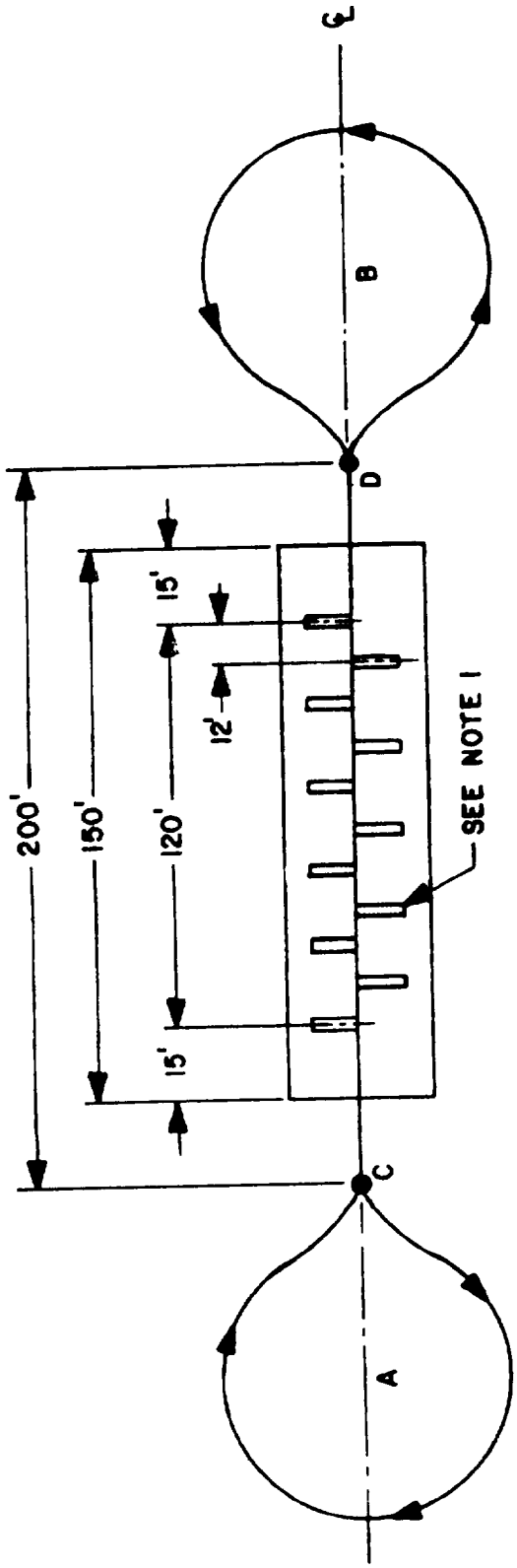


FIGURE A-3 TEST COURSE FOR EMERGENCY STEERING TEST X-2984

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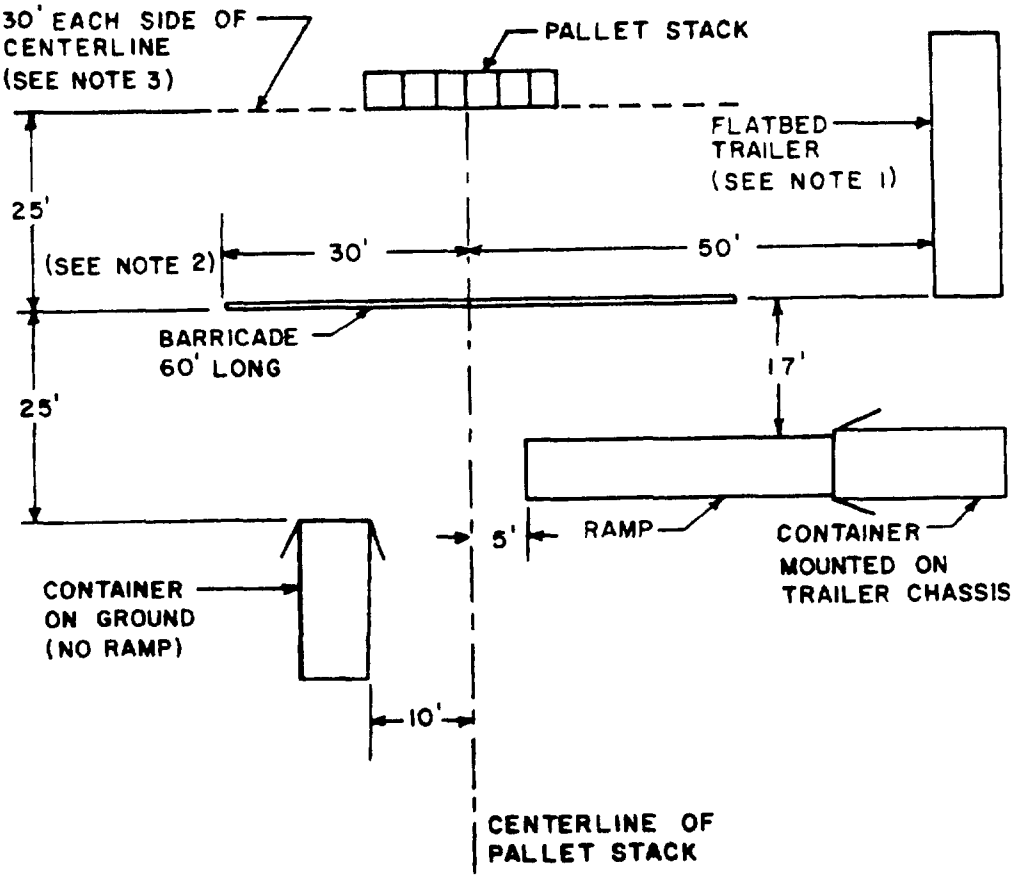
NOTES:

1. ELEVEN CROSSTIES, 6 FT. LONG, 12 FT. APART, ANCHORED TO PAVEMENT TO PREVENT MOVEMENT. CROSSTIES SHALL BE MINIMUM 4 INCHES HIGH X 4 INCHES WIDE.
2. SPEED SHALL NOT VARY BETWEEN POINTS C AND D.
3. TEST COURSE SHALL NOT BE PAVED EXCEPT FOR CROSSTIE AREA.

X-3351

FIGURE A-4. Typical durability course.

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NOTES:

1. ONE END OF FLATBED TRAILER ELEVATED TO PUT BED AT 8° ANGLE WITH THE HORIZONTAL.
2. ALL TRAVEL AROUND BARRICADE TO BE BETWEEN BARRICADE AND FLATBED TRAILER, EXCEPT WHEN APPROACHING RAMP TO ENTER TRAILER MOUNTED CONTAINER.
3. FORKLIFT SHALL NOT CROSS MARKED BOUNDARY.
4. PLATFORMS OR BOXES MAY BE USED TO DUPLICATE ACTUAL TRAILERS OR CONTAINERS.
5. TEST COURSE SHALL NOT BE PAVED.

FIGURE A-5. Container/trailer loading course.

X-3019A

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